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Module 12:

Shifting operations

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Ministerium für
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des Landes Nordrhein-Westfalen



Stand: 18. Februar 2026



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1 Introduction to switching operations

Switching operations are compensation processes that occur immediately after switching in electrical networks. As the term „electrical circuit“ suggests, switching plays an important role in electrical engineering. Even electrical networks in which no switching takes place are commonly referred to as circuits.

Switches are used in a variety of applications, including power electronics, information technology, communications and control engineering. After each effective switching operation, balancing processes occur, which can sometimes cause desirable or undesirable effects.

This module focuses on the investigation of these compensation processes.



Learning objectives: Introduction

The students learn:

- Recognise switching operations as non-stationary states following switching actions
- Understand switching operations in the context of other compensation processes
- Recognise challenges and opportunities associated with switching operations
- Distinguish between the switching behaviour of ideal and real switches

1.1 Settlement procedures

Compensation processes are generally processes in which a system strives to achieve a state of equilibrium after a disturbance.

A look at physics shows that balancing processes occur in many different ways in many areas. According to the second law of thermodynamics, a system always strives to achieve a state of equilibrium. Accordingly, every process can be regarded as a balancing process within the corresponding system boundaries. To a certain extent, switching processes can be described using analogies to other balancing processes.

Examples of compensation processes include heating a liquid, the oscillation of a pendulum, or the control process of a control loop.



Figure 1.1: Heating a liquid¹

¹Bartolomeo Pinelli, Excerpt from *A Peasant Family Cooking over a Campfire*, Licence CC0 1.0

1.2 Switching operations

Switching operations are defined as compensation operations immediately following switching. These are the main subject of the module, whereby ideal switches are always assumed.

Figure 1.2 shows an example of the time curve of a system variable $s(t)$ (e.g. current or voltage) during a switching operation. For comparison, two cases are shown for the connection of an excitation: one for a direct current (DC) and one for an alternating current (AC).

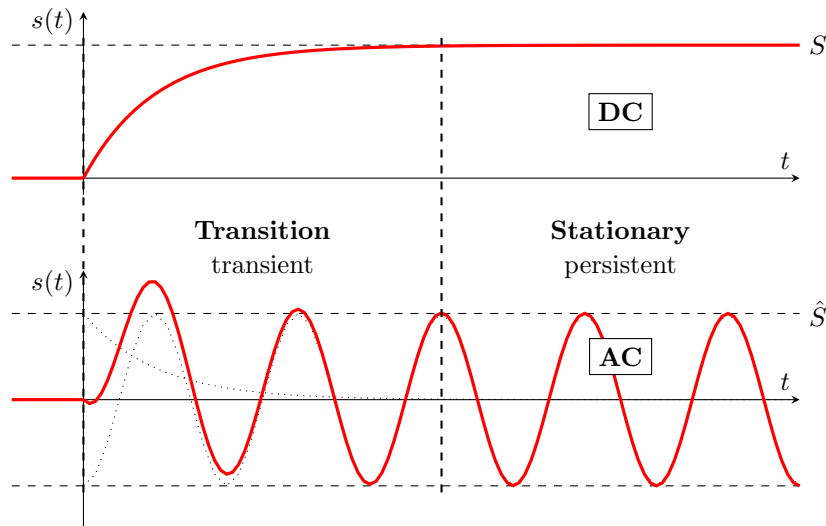


Figure 1.2: Comparison: AC and DC balancing process

The switching process shown begins with switching at $t = 0$ (dashed boundary, left) and ends when a steady state (persistent) is reached (dashed boundary, centre). The steady state of $s(t)$ corresponds to a constant value (DC) when excited with a constant value and a periodic value (AC) when excited with an alternating value.

The process shown in Fig. 1.2 during the transition (transient) from switching to reaching a steady state can also be referred to here as a balancing process (in general), a switching process (in particular) or a transient process (in particular).

Compensation processes during switching, as they occur in real switches, are not referred to as switching processes and are not examined in detail in this module. However, for the sake of completeness, a brief explanation can be found in section 1.4 when comparing ideal and real switches.

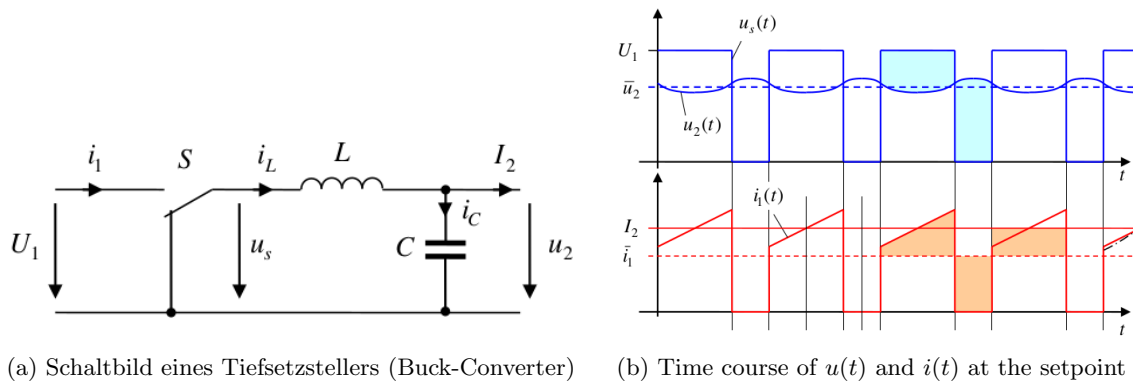
1.3 Examples of switching operations

In principle, switching operations occur every time a switch is made in electrical networks.

Typical examples are converters (power converters) in power electronics, which convert electrical energy into another form of electrical energy. By selectively switching semiconductor components (e.g. transistors, thyristors) the voltage, current or frequency can be changed. In each clock cycle, capacitances and inductances are alternately charged and discharged.

Figure 1.3a shows an example of a buck converter with a smoothing capacitor. This converts a DC voltage U_1 into a lower DC voltage U_2 . Figure 1.3b also shows the time curves of voltages and currents at the input and output of the buck converter.

<https://commons.wikimedia.org/w/index.php?curid=81414513>

Figure 1.3: Example: Switching operations on the step-down converter²

Without detailed calculation, it is apparent that the output voltage is not constant even in steady-state operation. The output voltage across the smoothing capacitor fluctuates because the capacitance discharges and recharges slightly in each switching period. The methods described in this module can be used to calculate and analyse such switching processes.

For a better understanding, however, simpler examples of switching processes without periodic switching are considered in Chapter 3. Other examples of switching processes are the switching on and off of devices, analogue-to-digital and digital-to-analogue conversion, and the charging of a capacitor by a bicycle light.

1.4 Comparison of ideal and real switches

For simplicity, this module always assumes ideal switches. To understand the limitations of this assumption, the difference between ideal and real switches is explained below. As shown in the circuit diagram in Figure 1.4, the switch voltage across a switch is considered. For this purpose, the switch is connected to a linear DC voltage source with voltage U_q .

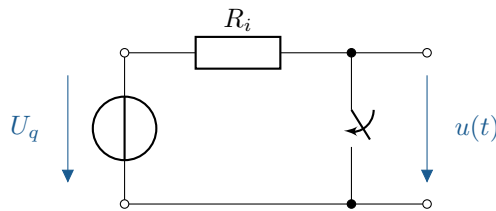


Figure 1.4: Circuit diagram, switch voltage

Figure 1.5 shows the switch voltage in forward and reverse operation and in the transition phases when opening or closing the switch. On the left is the voltage curve for an ideal switch, on the right for a real switch.

²Source: Circuit diagram and time sequence (modified, abridged): Joachim Böcker, GET2, Universität Paderborn

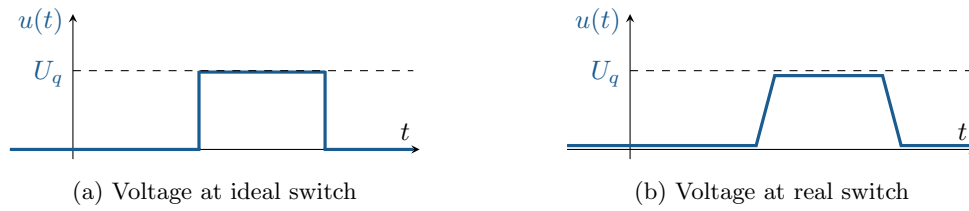


Figure 1.5: Comparison: switching sequences for ideal and real switches

An ideal switch changes instantaneously and without loss from a conductive state ($R = 0$) to a blocking state ($G = 0$) or vice versa as shown in Figure 1.5a. A real switch, on the other hand, does not conduct or block instantaneously as shown in a simplified form in Fig. 1.5b. Instead, latencies and losses occur. In forward mode, a small voltage is still present (forward resistance $R > 0$) and in reverse mode, a small current still flows (reverse conductance $G > 0$). This results in losses in the switch, especially during the transition phases when both voltage and current are present.

The latencies and losses of real switches result from their resistive, inductive and capacitive properties. Due to these properties, switching is always associated with compensation processes when viewed in real terms – within the switch. In practice, such effects are referred to as parasitic when undesirable.

Abbildung 1.6 zeigt exemplarisch einen MOSFET als Schalter zur Veranschaulichung. Zum Vergleich ist der MOSFET einmal ideale (ohne) und einmal real (mit kapazitiven Effekten) dargestellt.

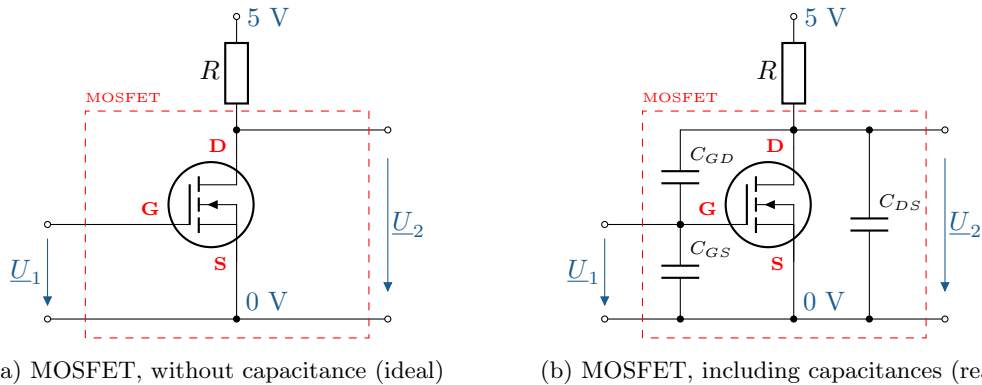


Figure 1.6: Comparison: Circuit with MOSFET, without (ideal) and with capacitive effects (real)

The capacitive effects between the terminals of the MOSFET cause a delay in switching, as the respective capacitances must first be charged or discharged during switching. The MOSFET example serves only to illustrate the differences between ideal and real switches. Compensation processes during switching operations are not considered further in this module. Switching operations always refer to the compensation processes after switching.

2 Fundamentals of calculating switching operations

Switching operations are calculated on the basis of differential equations (DGL).

Applying Kirchhoff's laws to circuits with energy storage devices leads to equations for voltages and currents in the respective components. This means that the respective instantaneous values of voltages and currents are not sufficient to describe the state of the system.

The time courses of voltages and currents during switching operations can be determined by solving their respective differential equations.

Learning objectives: Fundamentals

The students learn:

- Know and understand fundamental concepts (definitions)
- Be familiar with calculation methods in the time domain and frequency domain (Laplace)
- Set up differential equations for circuit quantities of LTI circuits
- Derive properties of differential equations from circuit properties
- Determine time responses during switching processes (solve differential equations)

2.1 Definitions

Table 2.1 provides an overview of important terms with abbreviated definitions used here. Certain terms are explained and discussed in more detail in the following sections. In particular, the definitions used here for transients, compensation, settling and switching processes

are explained in section 2.1.2 in contrast to the definitions found in other sources and the reasons for the choice of definitions used here are given.

The table serves as a reference for the use of terms in this module and can be used as a glossary.

2.1.1 Explanations

CONDITIONS:

Balanced: Describes a steady state after a balancing process.

Settled in: Describes a steady state immediately after a transient process. The term is also generally used for steady states after compensation processes in which no oscillation occurs and which are also independent of the direction of energy flow.

Stationary: State description for time-constant variables (e.g. with DC voltage supply) or time-periodic variables (e.g. with AC voltage) [2, p. 362]

SIZES:

The quantities **decay constant**, **damping constant**, **natural frequency**, **resonance frequency** and **time constant** are important parameters for describing the oscillatory behaviour of systems. They are explained in more detail in Chapter 3.3 using the example of an RLC series resonant circuit.

CASES:

The distinction between **aperiodic case**, **aperiodic limiting case**, **periodic case** in linear, time-invariant, oscillatory systems is explained in Chapter 3.3.1 using the example of an RLC series resonant circuit.

2.1.2 Discussion and delimitation

This section is not essential for understanding the module content and should be regarded as supplementary information. As some of the terms listed in Table 2.1 are used differently in the literature, the choice of definitions is justified and defined here.

PROCEDURES:

Table 2.1: Grouped terms with short definitions

G.	Terms	Definitions
General. Properties	Switch	„set to an (operating) state by pressing a switch“[1]
	Switch	Component, in this case for opening or closing an electrical connection
	Harmonic ¹	sinusoidal, describes an oscillation of the form: $y(t) = \hat{X} \cdot \sin(\omega t + \varphi)$
	Periodic ¹	„at regular intervals, regularly [occurring, recurring]“[1]
	Time-invariant ¹	invariant/unchanging over time
	Usually ²	Derivative(s) according to an independent variable (e.g. time t), cf. partial
	Homogeneous ²	Usually in the form: $\sum a_i(t) \frac{d^i}{dt^i} f_i(y(t)) \stackrel{!}{=} 0$ with interference function $b = 0$
	Linear ²	Usually in the form: $\sum a_i(t) \frac{d^i}{dt^i} y(t) = b(t)$ d.h. mit $\frac{d^i}{dt^i} y(t)$ linear
	Order ²	n [1] i.e. at most derivation of order n in DGL
Sizes	Decay constant	δ [Hz] Measure for the damping of linearly damped oscillatory systems
	Damping constant	δ [Hz] synonymous with decay constant
	Natural frequency	$f_{0/d}$ [Hz] Frequency at which a system can oscillate (undamped) on its own
	Resonance frequency	f_0 [Hz] Frequency at which a system resonates, equal to its natural frequency.
	Time constant	τ [s] Measure of steepness of exponential curves of the form $e^{-t/\tau}$
Procedures	Compensation process	Process in a system that strives for a steady state
	Oscillation process	Compensation process, oscillating, exergonic (spontaneous)
	Settling-in period	Compensation process, oscillating, endergon (not spontaneous)
	Switching process	non-steady state after switching
	Transient	Compensation process (synonym), cf. transient (adj.)
Condi.	Settled in	stationary after settling process
	Persistent	stationary from Latin <i>persistere</i> , German to remain, opposite of transient
	Stationary	Time-constant/periodic quantities (e.g. for DC/AC) [2, p. 362]
	Transient	transitional from Latin <i>transire</i> , German (to) cross over, opposite of persistent

¹generally²relating to DGL**Compensation process, m.**

- Weißgerber: Process from steady state after intervention to steady state
- Hagmann: Process from change to steady state
- here, based on Hagemann: Process when a specific steady state is desired

Like Hagmann, except that quasi „interrupted“ compensation processes are also referred to as compensation processes. This definition includes applications in power electronics such as step-up and step-down converters, assuming real components ($R > 0$).

Switching operation, m.

- Weißgerber: Compensation process after switching[3, p. 1]
- Hagmann: Synonym: compensation process, immediately after circuit change [sic!], transitions to steady state[2, p. 362]
- here: Compensation process triggered by switching (based on Hagmann)

In this module, the term ‘switching operation’ refers only to the process after (!) switching, which is triggered by this. The definition used here allows compensation processes that are independent of switching to be excluded.

Oscillation process, m.

Refers here to exergonic compensation processes, i.e. compensation processes that take place with positive net -energy dissipation. According to the literal meaning of the word, a settling process is the counterpart to a settling-in process.

Settling-in process, m.

Here, in the narrower sense, refers to endosmic compensation processes, i.e. compensation processes that take place with a positive net energy supply. Usually used here in the broader sense as a synonym for all compensation processes. Refers here, as usual, to both oscillatory and non-oscillatory processes. [Cf. the words oscillate, swing] The definition has been chosen to facilitate a more detailed classification of possible compensation processes. **Transient, w.**

Here considered synonymous with the process of compensation, freely translatable as „transition“ from Latin. Comparison transient (adj.) from Latin *transire* English (to) pass (through), opposite of persistent (adj.) from Latin *persistere* English to persist.

2.2 Calculation methods for switching operations

There are basically two methods available for calculating switching operations. Both methods are based on solving the differential equation (DE) of a circuit.

The DE is solved as follows:

- in the time domain (by decomposing into transient and steady states) **or**
- in the image domain (using Laplace transformation).

Figure 2.1 compares the calculation process in the time and image domains as a flowchart.

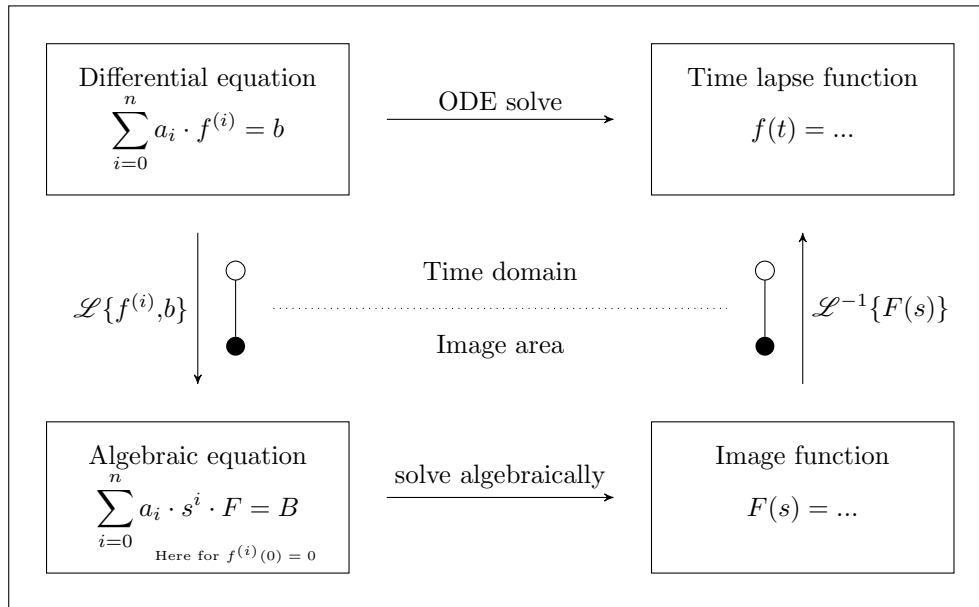


Figure 2.1: Flowchart, calculation methods for switching operations

The naming and presentation of both methods are based on [3, p. 51], but can also be found in a similar form in other textbooks on electrical engineering, control engineering and systems theory.

Solving the ODE in the time domain is the generally applicable method and is covered in detail in this module. The computational effort can vary greatly depending on the complexity of the ODE. The method is based on decomposing the system into a transient and a steady state.

Solving the ODE in the frequency domain using the Laplace transform is a special method that can be applied to linear, time-invariant systems with energy storage. It is suitable for vanishing initial conditions (see chapter 2.4). In some cases, the method can reduce the computational effort, as the transformed ODE can be solved algebraically. The method is not discussed further in this module.

The idea of switching to the image domain can be found again in Section 2.3 when setting up a differential equation and in Section 2.4.3 when applying complex AC calculation (see Module 7) to determine steady-state conditions.

2.3 Differential equations

The calculation of switching operations begins with the formulation of differential equations (ODG). The ODG for voltages and currents are based in principle on Kirchhoff's rules:

$$\text{Knot rule: } \sum_{j=1}^n \mathbf{i}_j = 0 \quad \text{mit } n \text{ Stromzweigen im Knoten} \quad (2.1)$$

$$\text{Mesh rule: } \sum_{k=1}^m \mathbf{u}_k = 0 \quad \text{mit } m \text{ Teilspannungen in Masche} \quad (2.2)$$

and the following current-voltage relationships (component equations) for R , L and C :

$$\text{Resistance: } u_R(t) = R \cdot \mathbf{i}_R(t) \quad (2.3)$$

$$\text{inductance: } u_L(t) = L \cdot \frac{d}{dt} \mathbf{i}_L(t) \quad (2.4)$$

$$\text{Capacity: } \mathbf{i}_C(t) = C \cdot \frac{d}{dt} u_C(t) \quad (2.5)$$

The most effective order in which to set up, transform and combine the equations depends on the respective circuit topology and the target variable.

Similar to solving differential equations (see section 2.2 and figure 2.1 on calculation methods), differential equations can be set up in both the time domain and the image domain.

The method in the time domain is the method discussed in more detail in this module. Figure 2.2 shows a schematic comparison of both approaches.

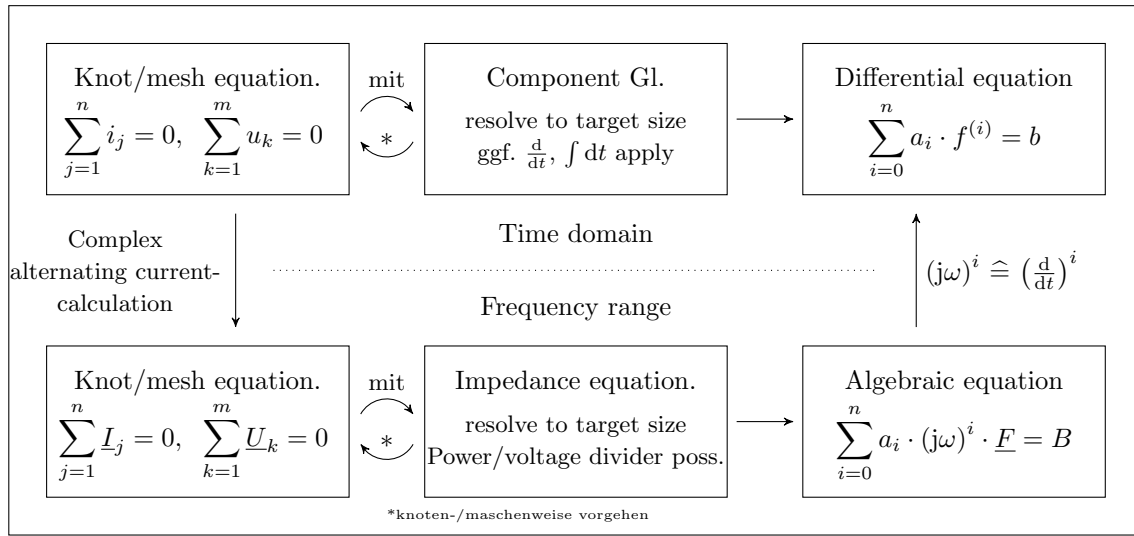


Abbildung 2.2: Flowchart, methods for setting up differential equations

For simple circuits with a single mesh (pure series connection) or two individual nodes (pure parallel connection), the differential equations can be derived directly from Kirchhoff's rules and the component equations (time domain). If necessary, the equation must be differentiated to obtain the required derivatives (or to remove integrals). For more complex networks, the computational effort is significantly greater.

One way to minimise the amount of calculation required is to use complex AC calculation methods (image area). This approach also works for DC circuits, as frequencies are not actually calculated. The aim is to set up the DGL by algebraically transforming the complex Kirchhoff's rules. Here, the circuit elements are replaced by impedances or admittances, and the quantity to be investigated is determined, for example, by a complex voltage divider. The transformation to the differential equation is performed by back-transforming the complex quantities into the time domain. $j\omega$ is replaced by $\frac{d}{dt}$.

The advantage of the approach in the image area is that complex voltage and current dividers can be used. This makes the approach particularly suitable for mixed series and parallel connections.

A similar approach would be the mesh flow method in matrix form with complex quantities. [See Module 7] However, this will not be considered further at this point.

2.3.1 Properties and general form

Only the following components are used in all of the circuits examined here: Capacitors, inductors, resistors, ideal voltage sources, ideal current sources, ideal electrical conductors and ideal switches. No other components such as transistors, diodes or operational amplifiers are used in the circuits.

To calculate switching operations, this module always assumes ideal components in isothermal, homogeneous, isotropic media. Other influences such as temperature dependencies, non-linear effects or similar factors are not taken into account.

The components listed are all **linear** and **time-invariant**. The same properties also apply to circuits that are composed solely of these components. This means that they are linear, time-invariant systems (LZI systems). Time invariance for switches applies with restrictions (in sections) for all time periods outside of switching times and without restrictions for all other components.

The following properties of the differential equations can be derived from the properties of the circuits:

- **ordinarily** (nur abhängig von einer Variable, hier der Zeit)
- **linear** (Linearität der Schaltungselemente)
- **const. coefficients** $a_i = konst.$ (Zeitinvarianz der Schaltungselemente)
- **inhomogeneous** $b(t) \neq 0$ (bei Anregung durch Spannungs- oder Stromquellen)
- **homogeneous** $b(t) = 0$ (ohne Anregung durch Spannungs- oder Stromquellen)

For a considered circuit variable y as a function of time t , this results in the following form for ordinary, inhomogeneous, linear ODEs with constant coefficients:

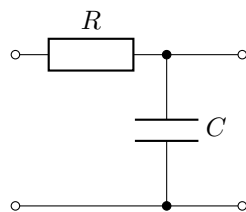
$$\sum_{i=0}^n a_i \cdot \frac{d^i y(t)}{dt^i} = b(t) \quad (2.6)$$

mit $a_i, b(t) \in \mathbb{R}; \quad i, n \in \mathbb{N}$

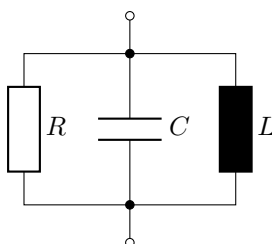
With the **order** n , the **disturbance function** $b(t)$ and the **constant coefficients** a_i .

2.3.2 Order of differential equations

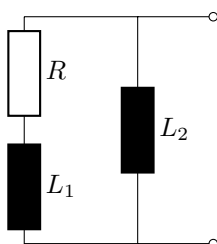
The structure of a circuit determines the form of the differential equations (DGLs) used to describe its system variables (currents, voltages). The order of the DGL corresponds directly to the number of energy storage devices that cannot be combined into a single energy storage device. [3, p. 4] Figure 2.3 shows several circuits and the order of the corresponding differential equations as examples.



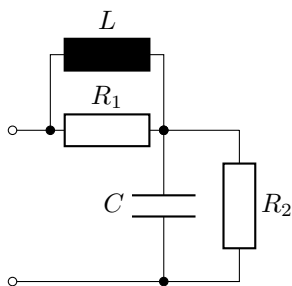
1 Energy storage $\rightarrow$ ODE 1st Ord.



2 Energy storage $\rightarrow$ ODE 2st Ord.



2 Energy storage $\rightarrow$ ODE 2st Ord.



2 Energy storage $\rightarrow$ ODE 2st Ord.

Figure 2.3: Example circuits and order of the differential equation

The order of the differential equation determines the number of constants in the general solution of the differential equation.

2.4 Calculation method in the time domain

The calculation of switching operations in the time domain is based on the solution of differential equations (ODEs) for the circuit variables. In general, there are an infinite number of solutions for ODEs.

Knowledge of the **initial conditions** (IC) is important for limiting the general solution. The IC are values of circuit variables at a specific point in time t_0 during the observation period (switching process). These are, for example, voltages or currents of individual components or their derivatives.

For ODEs of order n , n ABs are necessary to obtain a unique solution. If the ABs are not explicitly given, they can be derived from the circuit topology and the state immediately before switching, if necessary. A distinction must be made between which circuit variables can jump and which are continuous.

Since inductors and capacitors are energy storage devices, their behaviour at a point in time t depends on the energy supplied/removed up to that point.

In a brief comparison, with $E = \int p(t) dt = \int u(t) \cdot i(t) dt$ and Eqs. 2.4 and 2.5:

$$\textbf{Capacity:} \quad E_{el} = \frac{1}{2} C \cdot u^2 \quad \Rightarrow \quad u = \textit{steadily} \quad (\text{Electric en. storage}) \quad (2.7)$$

$$\textbf{Inductance:} \quad E_{mag} = \frac{1}{2} L \cdot i^2 \quad \Rightarrow \quad i = \textit{steadily} \quad (\text{Magnetic en. storage}) \quad (2.8)$$

Since energy can only move at a finite speed, **continuity** applies to u_C and i_L :

$$u_{C,t-} = u_{C,t+} \quad i_{L,t-} = i_{L,t+} \quad (2.9)$$

$$\text{mit} \quad \lim_{\pm\infty \rightarrow t} u_C(t) = u_{C,t\pm} \quad \lim_{\pm\infty \rightarrow t} i_L(t) = i_{L,t\pm} \quad (2.10)$$

This means that the voltage of a capacitance, just like the current of an inductance, is the same immediately before switching as it is at the beginning of the following switching process.

2.4.1 Decomposition of compensation processes and solution of inhomogeneous differential equations

The resulting time course of a variable $y(t)$ can be broken down during a compensation process. in a **volatile state** y_f und in a **steady state** y_e .

$$y(t) := y_f(t) + y_e(t) \quad (2.11)$$

The steady state y_e describes the stationary state that the variable y strives to reach. The transient state y_f , also known as the free state, represents the decaying part of y :

$$\lim_{t \rightarrow \infty} y_e(t) = \lim_{t \rightarrow \infty} y(t) \quad y_e = \text{stationary} \quad (2.12)$$

$$\lim_{t \rightarrow \infty} y_f(t) = 0 \quad y_f = y - y_e \quad (2.13)$$

Figure 2.4 shows an example of a compensation process with decomposition of a variable $y(t)$ into a transient and steady state. The graph corresponds to the voltage curve of a capacitor when charged with DC voltage across an ohmic resistor, as in Example 3.1.

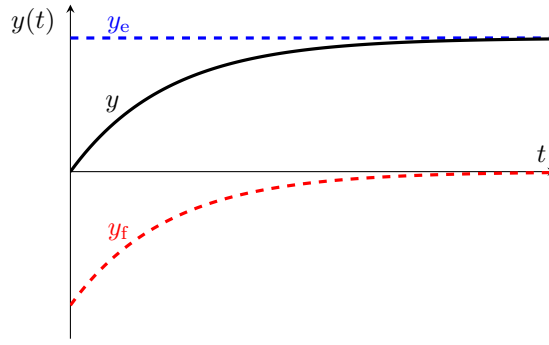


Figure 2.4: Compensation process broken down into steady state and transient state

Since the quantity $y(t)$ according to equation 2.12 transitions to $y_e(t)$ for $t \rightarrow \infty$, $y_e(t)$ must also satisfy the differential equation from equation 2.6 including the disturbance term b :

$$\sum_{i=0}^n a_i \cdot \frac{d^i(t)}{dt^i} y_e = b(t) \quad (2.14)$$

If we subtract the ODE for y_e from Eq. 2.14 from the general inhomogeneous ODE from Eq. 2.6, we obtain the homogeneous ODE for the transient state using the decomposition according to Eq. 2.11:

$$\sum_{i=0}^n a_i \cdot \frac{d^i(t)}{dt^i} y_f = 0 \quad (2.15)$$

Mathematically, the decomposition of the time course $y(t)$ into a transient and steady state corresponds to the decomposition of the **solution of the inhomogeneous ODE** y into the **solution of the homogeneous ODE** y_h (without disturbance term) and into a **particular solution** y_p of the inhomogeneous ODE with:

$$\text{homogeneous solution} \quad y_h = y_f \quad \text{volatile state} \quad (2.16)$$

$$\text{particular solution} \quad y_p = y_e \quad \text{settled state} \quad (2.17)$$

The breakdown of compensation processes into a steady state and a transient state can be found in a similar form in the textbooks by Albach[4], Hagmann[2] and Weißgerber[3].

2.4.2 Calculation of the homogeneous solution

For solving ordinary, linear, homogeneous differential equations with constant coefficients of the form:

$$\sum_{i=0}^n a_i \cdot \frac{d^i}{dt^i} y_h = 0 \quad (2.18)$$

the approach of an exponential function is generally used:

$$y_h(t) = K \cdot e^{\lambda t} \quad (2.19)$$

Here, K is a constant and λ is the solution of the characteristic polynomial and is also referred to as

the **eigenvalue**. By inserting the exponential approach in 2.18, differentiating and factoring out:

$$\begin{aligned} \sum_{i=0}^n a_i \cdot \frac{d^i}{dt^i} K \cdot e^{\lambda t} &= 0 \\ \left(\sum_{i=0}^n a_i \cdot \lambda^i \right) K \cdot e^{\lambda t} &= 0 \end{aligned} \quad (2.20)$$

the **characteristic polynomial** can be established:

$$\sum_{i=0}^n a_i \cdot \lambda^i = 0 \quad (2.21)$$

The maximum number of eigenvalues is given by the order of a differential equation. Since the coefficients a_i are real, the eigenvalues λ_i must also be real, or complex conjugate in pairs.

Condition: In order to satisfy the homogeneous differential equation in Equation 2.18, the real part of all eigenvalues must be negative $\text{Re}\{\lambda_i\} < 0$. Only then will the exponential terms from the approach in Equation 2.19 converge for $t \rightarrow \infty$.

The **general homogeneous solution** y_h of an n th-order ODE is the linear combination of n linearly independent solutions of the homogeneous ODE.

The form of the solution depends on whether there are distinct eigenvalues, identical eigenvalues (multiple zeroes of the characteristic polynomial) or complex conjugate eigenvalues in pairs.

If all eigenvalues λ_i are different, the homogeneous solution is:

$$y_h(t) = \sum_{i=1}^n K_i \cdot e^{\lambda_i t} \quad \text{für } \lambda_i \neq \lambda_j \quad \forall i, j \quad (2.22)$$

For complex conjugate eigenvalues in pairs, this solution can be reformulated. [Derivation in section 3.4.1.] For $\lambda_1 = \lambda_2^* = a + jb \in \mathbb{C}$, the homogeneous solution is:

$$\begin{aligned} y_h(t) &= K_1 \cdot e^{a \cdot t} \cdot \cos(b \cdot t) + K_2 \cdot e^{a \cdot t} \cdot \sin(b \cdot t) \\ &= C_0 \cdot e^{a \cdot t} \cdot \cos(b \cdot t + \varphi_0) \end{aligned} \quad (2.23)$$

If all eigenvalues λ_i are equal, the homogeneous solution is:

$$y_h(t) = \sum_{i=1}^n K_i \cdot t^{i-1} \cdot e^{\lambda t} \quad \text{für } \lambda = \lambda_i \quad \forall i \quad (2.24)$$

For mixed zeroes, the solution is obtained from a combination of the above solutions, as shown in Example 2.1.



Example 2.1: Linear independent solutions of a differential equation

This example is taken almost unchanged from [4, p. 241]. We are looking for the general solution to the following 5th order homogeneous differential equation:

$$\frac{d^5}{dt^5} y(t) + 7 \frac{d^4}{dt^4} y(t) + 26 \frac{d^3}{dt^3} y(t) + 62 \frac{d^2}{dt^2} y(t) + 8 \frac{d}{dt} y(t) + 75 y(t) = 0$$

The characteristic polynomial:

$$\lambda^5 + 7 \cdot \lambda^4 + 26 \cdot \lambda^3 + 62 \cdot \lambda^2 + 85 \cdot \lambda + 75 = 0$$

has one simple and two double, conjugate, complex zeros:

$$\lambda_1 = -3, \quad \lambda_2 = \lambda_3 = -1 + 2j, \quad \lambda_4 = \lambda_5 = -1 - 2j$$

Using the approaches from Eq. 2.22, 2.23 and 2.24, the general solution is:

$$y_h(t) = K_1 \cdot e^{-3t} + (K_2 + K_3 \cdot t) \cdot e^{-t} \sin(2t) + (K_4 + K_5 \cdot t) \cdot e^{-t} \cos(2t)$$

2.4.3 Calculation of the particular solution

Various methods can be chosen to calculate a particular solution y_p of an inhomogeneous ODE. One option is to choose an approach similar to the suggestion.

Specifically for switching operations, y_p corresponds to the steady state as derived in section 2.4.1. For linear, time-invariant switching elements, this implies:

- With DC excitation (constant quantities), the particular solution is also a constant quantity.
- With AC excitation (alternating quantities), the particular solution is also an alternating quantity.

The determination can be carried out using static network calculation methods. For this purpose, Kirchhoff's rules, the equations for the components R , L and C and the resulting current and voltage divider rules are sufficient. For alternating quantities, the complex alternating current equivalents of the methods listed are used.

Transformation of DGLs of sinusoidal quantities into the complex number space:

Since, in the case of AC excitation, the ODE 2.14 applies to the steady state for alternating quantities, the ODE can be transformed into the complex number space. The procedure corresponds to the transformation of the differential current and voltage relationships for R , L and C to their complex current and voltage relationships using their respective impedance $\underline{Z}$ and admittance $\underline{Y}$.

A sinusoidal quantity $y(t)$ with amplitude $\hat{Y}$ and a zero phase angle φ can be described with a complex amplitude pointer (phasor) $\underline{\hat{Y}}$ as follows [see Module 7]:

$$y(t) = \hat{Y} \cdot \cos(\omega t + \varphi) = \operatorname{Re} \left\{ \underline{\hat{Y}} \cdot e^{j\omega t} \right\} \quad \Longleftrightarrow \quad \underline{\hat{Y}} = \hat{Y} \cdot e^{j\varphi}$$

Derivatives with respect to time $\frac{d}{dt}$ are replaced by $j\omega$ in the complex number space:

$$\frac{d}{dt} y(t) = \hat{Y} \cdot \omega \cdot \cos(\omega t + \varphi + \frac{\pi}{2}) \quad \Longleftrightarrow \quad j\omega \cdot \underline{\hat{Y}} = \hat{Y} \cdot \omega \cdot e^{j(\varphi + \frac{\pi}{2})}$$

The factor $j\omega$ corresponds to the prefactor ω and a phase shift of $\pi/2$ when transformed back. This means that, in general, for differential equations of sinusoidal quantities (index ac) of the form in equation 2.6:

$$\sum_{i=0}^n a_i \cdot \left(\frac{d}{dt} \right)^i y_{ac} = b_{ac} \quad \Longleftrightarrow \quad \sum_{i=0}^n a_i \cdot (j\omega)^i \cdot \underline{\hat{Y}} = \underline{\hat{B}} \quad (2.25)$$

Alternatively, the imaginary part of complex amplitude vectors can also be used to represent sinusoidal alternating quantities. [Compare Module 7 – Periodic Quantities]

2.4.4 Summary

Calculating compensation processes in the time domain consists, on the one hand, of setting up a suitable differential equation and, on the other hand, of solving the differential equation that has been set up. Note box 22 summarises the calculation of compensation processes in the time domain in five steps. The steps are closely based on the procedure described by Weißgerber [3, p. 6].

Key point: Procedure for ODE

1.	Set up differential equation (lin., ord.)	$\sum a_i \cdot \frac{d^i y}{dt^i} = b$
2.	Transient state (homog. sol.)	$y_f = \sum K_i' \cdot e^{\lambda_i t}$
3.	Steady-state condition (part. sol.)	$y_e = y(t \rightarrow \infty)$
4.	Superposition (general sol.)	$y = y_f + y_e$
5.	Determine constant(s) (initial cond.)	$i_L, u_C = \text{continuous}$

* with $K_i' = \begin{cases} K_i & \text{for } \lambda_i \neq \lambda_j \quad \forall i, j \\ K_i \cdot t^k & \text{for } \lambda_i = \lambda_{i-k} \quad \text{with } k \in [0, 1, \dots, m-1] \end{cases}$

(simple NS)
(m-fold zero, k-th identical eigenvalue)

The order of steps 2 and 3 is, in principle, arbitrary. In mathematics, the order shown is more common with the determination of the general homogeneous solution before the solution of the particular solution(s). From an electrical engineering point of view, it may be advisable to first determine the steady state (particular solution).

3 Calculate switching operations in the time domain

This chapter examines switching processes in electrical circuits in the time domain. A distinction is made between switching processes for direct current and alternating current. The examination covers RC and RL elements, as well as RLC elements.

Learning objectives: Switching operations

The students learn:

- Calculate switching processes in the time domain
- Know and understand the switching behaviour of RC, RL and RLC elements
- Know the possibility of minimising switching processes with AC excitation
- Understand and analyse the oscillation behaviour of oscillating circuits

3.1 With direct voltage

This chapter calculates and illustrates the charging and discharging processes of capacitors and inductors in series with a resistor at DC voltage.

Switching processes with sinusoidal excitation are covered in Chapter 3.2.

3.1.1 In terms of capacity

Example 3.1 shows the calculation method for determining the voltage curve $u_C(t)$ of a capacitance C that is initially discharged when charging with DC voltage for $t \geq 0$ across a resistor R and the time curve of the charging current $i(t)$. The calculation method used corresponds to the procedure for ODE in the time domain according to note 22.

The results show the exponential convergence of $u_C(t)$ to the excitation U_q :

$$u_C(t) = U_q \cdot \left(1 - e^{-t/\tau}\right) \quad (3.1)$$

and the exponential decay of $i(t)$ and $u_R(t)$ respectively from their maximum values U_q/R and U_q :

$$i(t) = \frac{U_q}{R} \cdot e^{-t/\tau} \quad (3.2)$$

$$u_R(t) = U_q \cdot e^{-t/\tau} \quad (3.3)$$

until the capacitance is charged. The **time constant** τ determines the steepness of the exponential curves. For the RC element, $\tau = RC$. The unit for τ is the same as that for time.

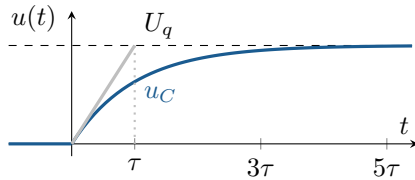


Figure 3.1: Exp. Approximation and τ

Table 3.1: Typical values Time constant τ

t	τ	3τ	5τ
$1 - e^{-t/\tau}$	63,2%	95,0%	99,3%
$e^{-t/\tau}$	36,8%	5,0%	0,7%

Rule of thumb: „Steady state“ after 5τ

As shown in Figure 3.1, the tangent of u_C intersects the terminal value U_q at the point $u_C(t = 0)$ at the time $t = \tau$. In general, at every point of exponential approximation/decline, the tangent reaches the terminal value after τ .

Table 3.1 shows typical values for exponential approximations/declines after multiples of τ . The values are relative to the maximum value (starting or target limit value). Such a curve is considered to be „stabilised“ after 5τ .

Example 3.2 shows the calculation method for determining the voltage curve $u_C(t)$ of a capacitance C when discharging across a resistor R and the corresponding time curves. Similar to the charging process, this results in exponential curves with the same time constant but with different start and end values.

The calculation method in examples 3.2 and 3.2 is essentially identical. The only differences between the charging and discharging processes are the initial condition $u_C(0)$ (at switching time $t = 0$) and the boundary condition $u_{C,e}$ (for $t \rightarrow \infty$). This results in different particular solutions and different constants K in the general solution.

In general terms, the voltage curve u_C of a capacitance C during charging/discharging with DC

voltage across a resistor R from a switching point t_0 can be described by:

$$u_C(t - t_0) = \underbrace{u_{C,e}}_{u_{C,p}} + \underbrace{(u_C(t_0) - u_{C,e})}_{u_{C,h}} \cdot e^{-\frac{t-t_0}{\tau}} \quad (3.4)$$

Rewritten slightly, the voltage curve can also be described quite clearly as follows:

$$u_C(t - t_0) = \underbrace{u_C(t_0)}_{\text{Initial value}} + \underbrace{(u_{C,e} - u_C(t_0))}_{\text{Difference}} \cdot \underbrace{\left(1 - e^{-\frac{t-t_0}{\tau}}\right)}_{\text{Exp. approximation}} \quad (3.5)$$

The general solution in equation 3.4 or 3.5 only applies to excitations of equal magnitude. The solution is typical for lossy, first-order linear systems (with an energy storage device), which is why it also occurs in a similar form in RL elements.

Digression: First-order ODE and time constant τ

The exponential form is typical for first-order linear systems with energy storage and results from the solution of the linear, ordinary first-order differential equation with constant coefficients. By transformation ($a_0 = 1$), the time constant τ can be read directly from the differential equation ($a_1 = \tau$):

$$\tau \cdot \frac{d}{dt} y(t) + y(t) = b(t) \quad (3.6)$$

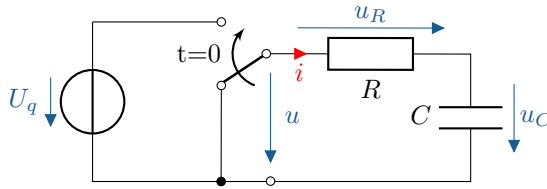
This allows the homogeneous solution to be specified directly with:

$$y_h(t) = K \cdot e^{-\frac{t}{\tau}} \quad (3.7)$$

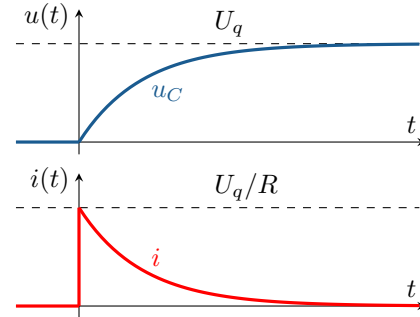
The time constant is a measure of the speed at which a system responds to a change.

Example 3.1: RC element DC charging process

Determination of voltage u_C and charging current i of capacitance C when charging via resistor R with DC voltage U_q for $t \geq 0$. The capacitance is discharged at the start.



given: $i(0), U_q, R, C$
 wanted: $i(t), u_C(t)$



1.) Set up ODE: von $u_C(t)$ for $t > 0$

Based on the mesh equation, u_R is expressed as a function of u_C :

$$\begin{aligned} \text{Mesh :} \quad & u_R + u_C = U_q && \text{with } u_R = R \cdot i \\ & R \cdot i + u_C = U_q && \text{with } i = C \cdot \frac{d}{dt} u_C \\ \text{ODE :} \quad & R \cdot C \cdot \frac{d}{dt} u_C + u_C = U_q \end{aligned}$$

2.) Homogeneous solution, gen.: (fleeting) (without interference)

$$\begin{aligned} \text{ODE}_h : \quad & RC \cdot \frac{d}{dt} u_{C,h} + u_{C,h} = 0 && \text{Approach: } u_{C,h} = K \cdot e^{\lambda t} \\ & K \cdot (RC \cdot \lambda e^{\lambda t} + e^{\lambda t}) = 0 \\ \text{char. Pol.:} \quad & RC \cdot \lambda + 1 = 0 && \Rightarrow \lambda = -\frac{1}{\tau} = -\frac{1}{RC} \\ \text{hom. Solu. :} \quad & u_{C,h} = K \cdot e^{-t/\tau} && \text{with } \tau = RC \end{aligned}$$

3.) Particular solution: (settled in) (for $t \rightarrow \infty$)

$$\begin{aligned} \text{ODE}_p : \quad & \cancel{RC \cdot \frac{d}{dt}} u_{C,p} + u_{C,p} = U_q && \text{with } u_{C,p} = \text{const.} \\ \text{part. Lsg. :} \quad & u_{C,p} = U_q \end{aligned}$$

4.) Inhomogeneous solution, gen.: (Superposition)

$$\begin{aligned} & u_C(t) = u_{C,h}(t) + u_{C,p}(t) && u_C(t) = u_{C,f}(t) + u_{C,e}(t) \\ \text{general sol.:} \quad & u_C(t) = K \cdot e^{-t/\tau} + U_q \end{aligned}$$

5.) Determine constants: (Set initial conditions)

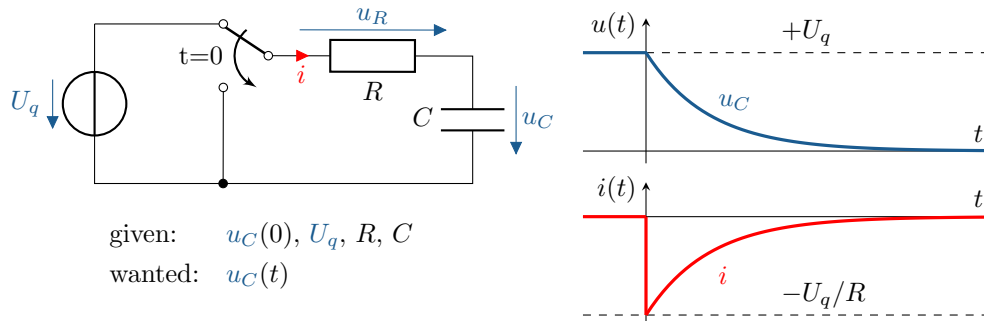
$$\begin{aligned} \text{initial condition} \quad & u_C(0) = K + U_q \stackrel{!}{=} \frac{0}{1} && \Rightarrow K = -U_q \\ & u_C(t) = -U_q \cdot e^{-t/\tau} + U_q \\ \text{special general solu.:} \quad & u_C(t) = U_q \left(1 - e^{-t/\tau}\right) \end{aligned}$$

Determine current i from voltage u_C :

$$\text{Current:} \quad i = \frac{U_q}{R} \cdot e^{-t/\tau} \quad \text{with } i = C \cdot \frac{d}{dt} u_C$$

Example 3.2: RC element discharge process

Determination of voltage u_C and current i_C of a capacitance C when discharging through a resistor R for $t \geq 0$. The capacitance is initially charged to U_q .



given: $u_C(0), U_q, R, C$
 wanted: $u_C(t)$

1.) **Set up ODE:** from $u_C(t)$ for $t > 0$, analogous to example 3.1

$$\begin{aligned} \text{Mesh :} \quad & u_R + u_C = 0 & \text{with } u_R &= R \cdot i \\ \text{ODE :} \quad & RC \cdot \frac{d}{dt} u_C + u_C = 0 & \text{with } i &= C \cdot \frac{d}{dt} u_C \end{aligned}$$

2.) **Homogeneous solution, gen.:** (fleeting) Identical to example 3.1

$$\text{hom. Lsg. :} \quad u_{C,h} = K \cdot e^{-t/\tau} \quad \text{with } \tau = RC$$

The particular solution $u_{C,p}$ is omitted, since u_C becomes zero when $t \rightarrow \infty$.

3. **Part. Lsg.:** (settled in) and 4. **Superposition:**

$$\text{general sol.:} \quad u_C = u_{C,h} + \cancel{u_{C,p}} \quad \text{with } u_{C,p} = 0$$

5.) **Determine constants:** (Set initial conditions)

$$\begin{aligned} \text{Initial condition.} \quad & u_C(0) = K \cdot e^0 \stackrel{!}{=} U_q & \Rightarrow & K = U_q \\ \text{special general sol.:} \quad & u_C(t) = U_q \cdot e^{-t/\tau} \end{aligned}$$

Determine current i from voltage u_C :

$$\text{Current:} \quad i = -\frac{U_q}{R} \cdot e^{-t/\tau} \quad \text{with } i = C \cdot \frac{d}{dt} u_C$$

3.1.2 For Inductances

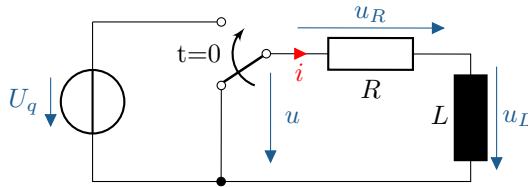
The calculation method for determining the current or voltage curve for inductors when excited with DC voltage is identical to the calculation for capacitors in Chapter 3.1.1. The resulting time behaviour with exponential curves and time constants is analogous to that described there.

Examples are given in Example 3.3 and 3.4 the charging current $i(t)$ and the inductance voltage $u_L(t)$ of an inductance L during charging via a resistor R with DC voltage U_q are calculated and displayed for $t \geq 0$.

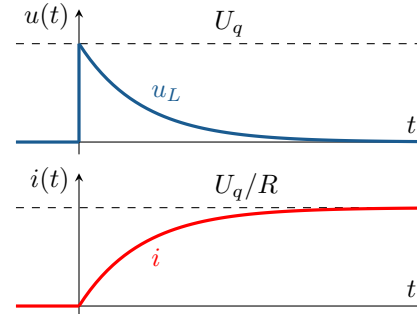
The examples differ only in the calculation method. In example 3.3, the differential equation for $i(t)$ is set up and solved, and in example 3.4, the differential equation for $u_L(t)$ is set up and solved.

Example 3.3: RL element DC charging process - Via ODE from i

Determination of charging current i and voltage u_L of inductance L when charging via resistor R with DC voltage U_q from switch-on time $t = 0$. No current flows before this point.



given: $i(0), U_q, R, L$
 wanted: $u_L(t), i(t)$



1.) Set up ODE: von $i(t)$ for $t \geq 0$

Based on the mesh equation, u_R and u_L are rewritten according to i :

$$\begin{aligned} \text{Mesh :} \quad & u_R + u_L = U_q && \text{with } u_R = R \cdot i \\ & R \cdot i + L \cdot \frac{d}{dt} i = U_q && \text{with } u_L = L \cdot \frac{d}{dt} i \\ \text{ODE :} \quad & \frac{L}{R} \cdot \frac{d}{dt} i + i = \frac{U_q}{R} \end{aligned}$$

2.) Homogeneous solution, gen.: (fleeting) (without interference)

$$\begin{aligned} \text{ODE}_h : \quad & \frac{L}{R} \cdot \frac{d}{dt} i_h + i_h = 0 && \text{Approach: } i_h = K \cdot e^{\lambda t} \\ \text{char. Pol.:} \quad & \frac{L}{R} \cdot \lambda + 1 = 0 && \Rightarrow \lambda = -\frac{1}{\tau} = -\frac{R}{L} \\ \text{hom. sol. :} \quad & i_h = K \cdot e^{-t/\tau} && \text{with } \tau = \frac{L}{R} \end{aligned}$$

3.) Particular solution: (settled in) with $t \rightarrow \infty$

$$\begin{aligned} \text{ODE}_p : \quad & \cancel{\frac{L}{R} \cdot \frac{d}{dt} i_p} + i_p = \frac{U_q}{R} && \text{with } i_p = \text{const.} \\ \text{part. sol. :} \quad & i_p = \frac{U_q}{R} \end{aligned}$$

4.) Non-homogeneous solution, gen.: (superposition)

$$\text{general sol.:} \quad i(t) = \underbrace{K \cdot e^{-t/\tau}}_{i_h} + \underbrace{\frac{U_q}{R}}_{i_p} \quad \text{with } i_h = i_f, \quad i_p = i_e$$

5.) Determine constants: (Apply initial conditions)

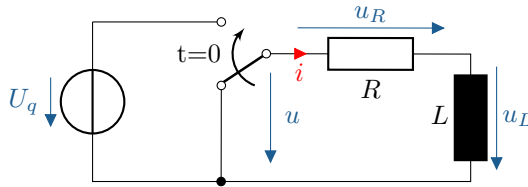
$$\begin{aligned} \text{Initial condition} \quad & i(0) = K \cdot e^0 + \frac{U_q}{R} \stackrel{!}{=} 0 && \Rightarrow K = -\frac{U_q}{R} \\ \text{clear sol.:} \quad & i(t) = \frac{U_q}{R} \cdot \left(1 - e^{-t/\tau}\right) \end{aligned}$$

Determine voltage u_L from current i :

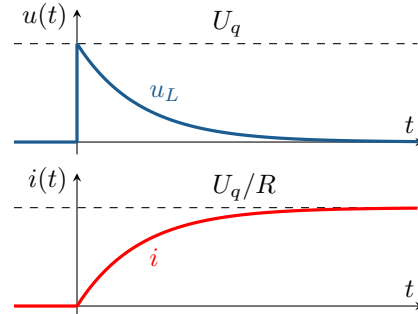
$$\text{Voltage:} \quad u_L = U_q \cdot e^{-t/\tau} \quad \text{with } u_L = L \cdot \frac{d}{dt} i$$

Example 3.4: RL element DC charging process - Via ODE of u_L

Determination of voltage u_L and charging current i of inductance L when charging via resistor R with DC voltage U_q for $t \geq 0$. The circuit is interrupted before $t = 0$.



given: $i(0), U_q, R, L$
 wanted: $u_L(t), i(t)$



1.) Set up ODE: from $u_L(t)$ for $t \geq 0$

Based on the mesh equation, u_R is expressed as a function of u_L :

$$\begin{aligned} \text{Mesh :} \quad & u_R + u_L = U_q && \text{with } u_R = R \cdot i \\ & R \cdot i + u_L = U_q && \left| \cdot \frac{d}{dt} \right. \\ & R \cdot \frac{u_L}{L} + \frac{d}{dt} u_L = \frac{d}{dt} U_q && \text{with } u_L = L \cdot \frac{di}{dt} \\ \text{ODE :} \quad & \frac{L}{R} \cdot \frac{d}{dt} u_L + u_L = 0 \end{aligned}$$

2.) Homogeneous solution, gen.: (fleeting) (without interference)

$$\begin{aligned} \text{ODE}_h : \quad & \frac{L}{R} \cdot \frac{d}{dt} u_{L,h} + u_{L,h} = 0 && \text{Approach: } u_{L,h} = K \cdot e^{\lambda t} \\ \text{char. Pol.:} \quad & \frac{L}{R} \cdot \lambda + 1 = 0 && \Rightarrow \lambda = -\frac{1}{\tau} = -\frac{R}{L} \\ \text{hom. sol. :} \quad & u_{L,h} = K \cdot e^{-t/\tau} && \text{with } \tau = \frac{L}{R} \end{aligned}$$

3.) Particular sol.: (settled in) with $t \rightarrow \infty$

$$\begin{aligned} \text{ODE}_p : \quad & \frac{L}{R} \cdot \frac{d}{dt} u_{L,p} + u_{L,p} = 0 && \text{with } u_{L,p} = \text{const.} \\ \text{part. sol. :} \quad & u_{L,p} = 0 \end{aligned}$$

4.) Inhomogeneous solution, gen.: (superposition)

$$\begin{aligned} u_L(t) &= u_{L,h}(t) + u_{L,p}(t) && u_L(t) = u_{L,f}(t) + u_{L,e}(t) \\ \text{clear sol.:} \quad & u_L(t) = K \cdot e^{-t/\tau} \end{aligned}$$

5.) Determine const.: (Set initial conditions)

$$\begin{aligned} \text{Initial condition.} \quad & i(0) = \frac{u_R(0)}{R} \stackrel{!}{=} 0 && \Rightarrow u_L(0) = U_q - u_R(0) \\ & u_L(0) = K \cdot e^{-0} \stackrel{!}{=} U_q && \Rightarrow K = U_q \\ \text{clear sol.:} \quad & u_L(t) = U_q \cdot e^{-t/\tau} \end{aligned}$$

Determine current i from voltage u_L :

$$\text{Current:} \quad i = \frac{U_q}{R} \cdot \left(1 - e^{-t/\tau}\right) \quad \text{with } i = \frac{u_R}{R} = \frac{U_q - u_L}{R}$$

The discharge process of an inductance L via a resistor R is similar to the charging process via a resistor R with a DC voltage U_q . The short circuit across R corresponds to a voltage source with $U_q = 0$ V. The discharge process and the charging process differ only in the initial conditions and the disturbance term of the respective differential equation. The time constant τ is identical for the charging and discharging processes. A detailed calculation is therefore not provided here.

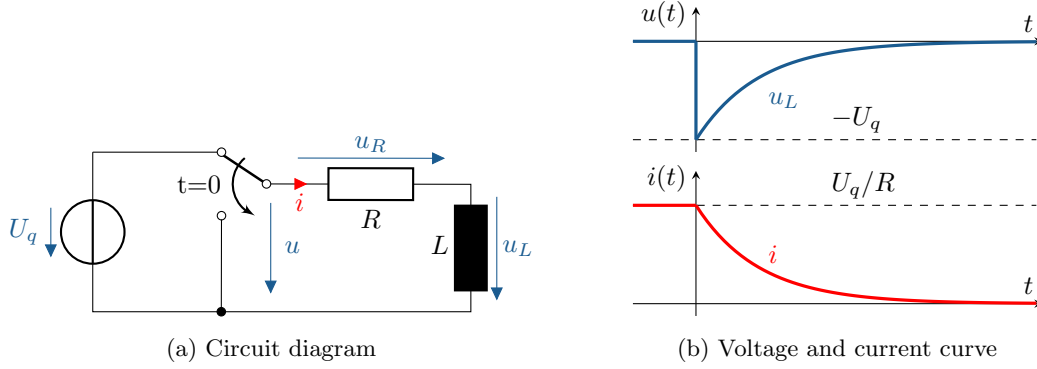


Figure 3.2: RL element, discharge process

Figure 3.2 shows the circuit diagram (a) and the voltage and current curve (b) of the inductance L when discharging via a resistor R from switching time $t = 0$. Before switching, the inductance is connected in series with the resistor R to a DC voltage source U_q . At the switching point, the inductance is fully charged (steady state) and conducts the constant current U_q/R before switching.

Real inductances can be modelled by a series connection of an ideal inductance L and an ohmic resistor R_{Cu} , which corresponds to the copper losses. connecting an ohmic resistor R_{Fe} in parallel directly to the inductance L , iron losses can also be modelled.

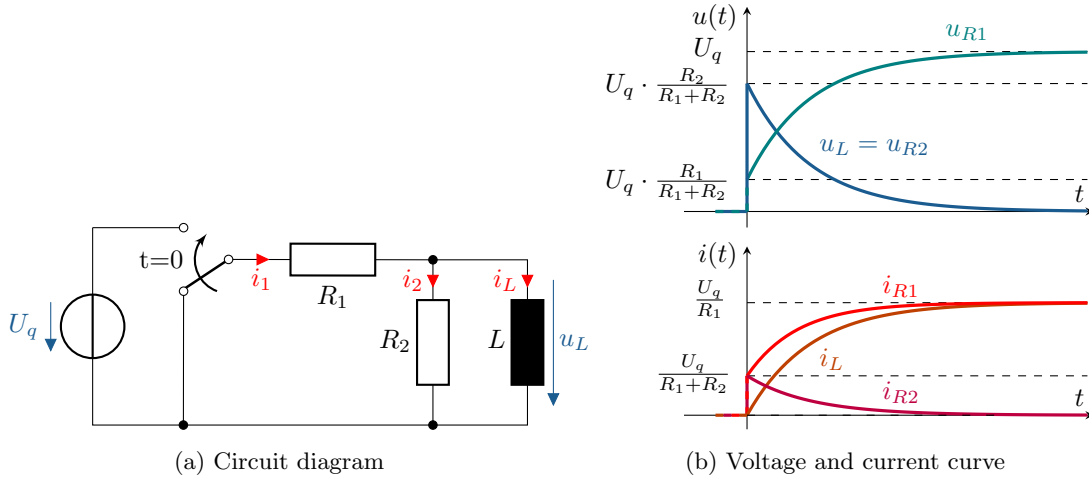


Figure 3.3: RL element, real inductance, charging process at ideal voltage source

Figure 3.3 shows the circuit diagram (a) of a real coil with inductance L in series with a resistor R_{Cu} (here R_1). Immediately parallel to the inductance is the ohmic resistor R_{Fe} (corresponding to R_2). The inductance is charged from $t = 0$ via an ideal constant voltage source U_q and is completely discharged beforehand. The corresponding voltage and current curves (b) are shown alongside.

Compared to the charging process of a pure RL series connection in Example 3.4, the voltage across the inductance does not jump to the full source voltage but to a lower value limited by the vol-

tage divider $\frac{R_2}{R_1+R_2}$. Due to iron losses, the input current jumps at the switching point and then exponentially approaches the current $\frac{U_q}{R_1}$. The current through R_2 jumps to the value $\frac{U_q}{R_1+R_2}$ when switching and then drops exponentially. The current through the inductance approaches the current $\frac{U_q}{R_1}$ exponentially. The time constant is $\tau = \frac{L}{R_1 \parallel R_2}$.

3.2 With sinusoidal excitation

The procedure for calculating switching operations with sinusoidal excitation is essentially the same as the procedure for uniform excitation. Therefore, this chapter only deals with the special features of sinusoidal excitation and explains the calculation using an example.

The differential equations for circuits with alternating voltage or current sources differ from those for direct voltage or current sources only in terms of the right-hand side of the differential equation, the disturbance term. The disturbance term results from the excitation and is also sinusoidal in the case of sinusoidal excitation of linear, time-invariant systems.

Due to the time dependency of sinusoidal excitation, the exact switching point is decisive for the behaviour of the system. More precisely, the phase position of the excitation at the switching point is decisive.

Using the example of an RC element, Chapter 3.2.1 shows how to calculate the switching process for sinusoidal excitation. It explains in detail how the switching process depends on the switching time and the possibility of transient peak currents and voltages.

Example 3.5 summarises the calculation of a switching operation for an RL element when excited with alternating voltage.

3.2.1 In terms of capacity

To explain the special features of switching processes with sinusoidal excitation, we consider the example of an RC circuit. The calculation is performed as in Example 3.1 according to Rule 22.

A capacitance C is connected via a resistor R from the switching time $t = t_0$ to an ideal AC voltage source with voltage $u_q(t) = \hat{U}_q \cdot \cos(\omega t + \varphi_q)$. The capacitance C is completely discharged at the start. Figure 3.4 shows the circuit diagram for this arrangement.

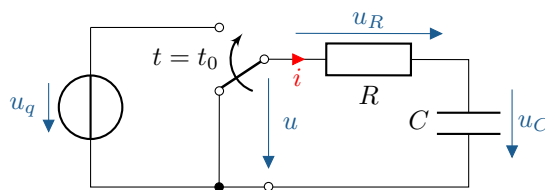


Figure 3.4: Circuit diagram, RC element (series) after switching to ideal AC voltage source

1.) Set up a differential equation:

The inhomogeneous ODE of u_C for $t \geq t_0$ is obtained analogously to Example 3.1 as:

$$RC \cdot \frac{d}{dt} u_C(t) + u_C(t) = u_q(t) \quad (3.8)$$

$$\text{mit } u_q(t) = \hat{U}_q \cdot \cos(\omega t + \varphi_q) \quad (3.9)$$

Compared to example 3.1, only the disturbance term has been changed. The DC voltage term U_q has been replaced here by the AC voltage term u_q .

2.) Homogeneous solution, general (fleeting state):

The homogeneous solution $u_{C,h}$ is independent of the disturbance term and thus of the excitation. The solution is taken from Example 3.1, taking into account the variable switching time t_0 :

$$\begin{aligned} u_{C,h} &= K \cdot e^{-\frac{t'}{\tau}} \quad \text{with } t' = t - t_0 \\ &= K \cdot e^{-\frac{t-t_0}{\tau}} \quad \text{with } \tau = RC \end{aligned} \quad (3.10)$$

Since the exponential approach only applies from switching time t_0 , this is subtracted from time t .

3.) Particular solution (steady state):

The particular solution $u_{C,p}$ also corresponds to a sinusoidal quantity with the same frequency as the excitation in the case of sinusoidal excitation, as described in Chapter 2.4.3. The steady state can be described using the complex amplitude indicator $\underline{\hat{U}}_{C,p}$ as follows:

$$u_{C,p} = \hat{U}_{C,p} \cdot \cos(\omega t + \varphi_{C,p}) \quad (3.11)$$

$$= \operatorname{Re} \left\{ \underline{\hat{U}}_{C,p} \cdot e^{j\omega t} \right\} \quad \text{with } \underline{\hat{U}}_{C,p} = \hat{U}_{C,p} \cdot e^{j\varphi_{C,p}} \quad (3.12)$$

The amplitude $\hat{U}_{C,p}$ and the (load) phase angle $\varphi_{C,p}$ can be determined according to complex voltage dividers.

$$\begin{aligned} \underline{\hat{U}}_{C,p} &= \underline{\hat{U}}_q \cdot \frac{\underline{Z}_C}{R + \underline{Z}_C} = \underline{\hat{U}}_q \cdot \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \\ &= \frac{\underline{\hat{U}}_q}{1 + j\omega RC} \quad \text{with } \underline{\hat{U}}_q = \hat{U}_q \cdot e^{j\varphi_q} \end{aligned} \quad (3.13)$$

For the amplitude $\hat{U}_{C,p}$ and the phase $\varphi_{C,p}$, the following follows from Eq. 3.13:

$$\hat{U}_{C,p} = \frac{\hat{U}_q}{\sqrt{1 + (\omega RC)^2}} \quad \text{und} \quad \varphi_{C,p} = \varphi_q - \arctan(\omega RC) \quad (3.14)$$

4.) Inhomogeneous solution, generally (superposition):

The superposition of Eq. 3.10 and 3.11 yields the general solution of the inhomogeneous differential equation:

$$\begin{aligned} u_C &= u_{C,h} + u_{C,p} \\ &= K \cdot e^{-\frac{t-t_0}{\tau}} + \hat{U}_{C,p} \cdot \cos(\omega t + \varphi_{C,p}) \end{aligned} \quad (3.15)$$

5.) Determine constants (Initial condition):

From the initial condition that the capacitance is discharged at the start, the following applies to the constant K :

$$\begin{aligned} u_C(t_0) &\stackrel{!}{=} 0 \\ K \cdot e^{\cancel{0}} + \hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,p}) &= 0 \\ K &= -\hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,p}) \end{aligned} \quad (3.16)$$

Using Eq. 3.16 inserted into 3.15, we obtain the unique solution for $u_C(t)$:

$$u_C(t) = \underbrace{-\hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,p}) \cdot e^{-\frac{t-t_0}{\tau}}}_{u_{C,h}} + \underbrace{\hat{U}_{C,p} \cdot \cos(\omega t + \varphi_{C,p})}_{u_{C,p}} \quad (3.17)$$

Equation 3.17 shows that the transient state $u_{C,f} = u_{C,h}$ tends exponentially towards zero from t_0 . In contrast to the DC case (example 3.1), the starting value of the exponential curve K from equation 3.16 depends on t_0 , since the steady state $u_{C,e} = u_{C,p}$ is time-dependent.

Behaviour when changing the switching point/time constant:

Figure 3.5a shows the voltage curve of the excitation u_q at a zero phase angle φ_q of zero and the voltage curve of the capacitance u_C . For comparison, u_C is shown for two different switching times t_0 . One is at the negative peak value and the other is at the zero crossing of $u_{C,p}$. u_C is shown as a superposition of the transient state $u_{C,h}$ (black, dashed) and the steady state $u_{C,p}$ (black).

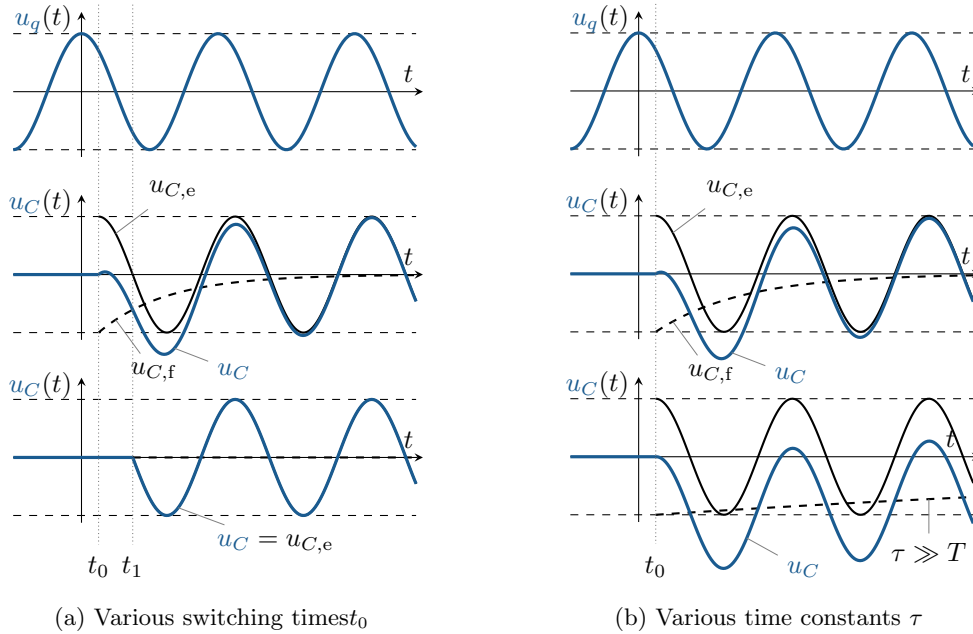


Figure 3.5: Voltage curve u_C for RC circuit after switching to AC voltage source u_q

As can be seen, the compensation process is maximal ($|K| = \max$) when the difference $|u_{C,p}(t_0) - u_C(t_0)|$ is maximal. If $u_{C,p}$ already satisfies the initial condition with $u_{C,p}(t_0) \stackrel{!}{=} u_C(t_0)$, the transient state is completely eliminated $K = 0 \Rightarrow u_{C,h} = 0$ and no compensation process occurs.

Figure 3.5b shows the voltage curve u_C for different time constants τ . In one case, $\tau = \frac{2\pi}{\omega}$, and in the other, $\tau = \frac{20\pi}{\omega}$. In both cases, t_0 is chosen so that the equalisation process is maximised.

With large time constants, the transient state decays more slowly, which can lead to overvoltages/overcurrents at unfavourable switching times. If u_C is discharged at the start according to the initial condition in the example ($u_C(t_0) = 0$), u_C can rise to twice the peak value of $u_{C,p}$. If, on the other hand, u_C is already charged to the negative/positive peak value of $u_{C,p}$ at the start, u_C can even rise to three times the peak value of $u_{C,p}$.



Key point: Compensation processes with AC excitation

Skilful switching can prevent compensation processes in the case of AC excitation.

With large time constants, overvalues of up to twice/three times the peak value are possible.

3.2.2 For Inductors

Example 3.5 shows an example of the calculation of a switching operation for an RL element when excited with alternating voltage according to Merksatz 22.

The calculation of switching processes for inductors with sinusoidal excitation is analogous to the calculation for capacitors in Chapter 3.2.1. The same special features apply with regard to switching times and possible transient peak currents and voltages as for capacitors.

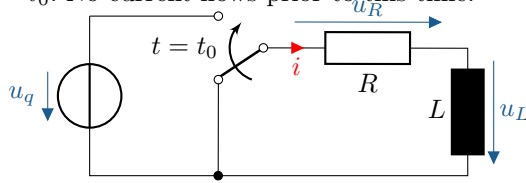
In the example, the DGL of the particular solution is transformed according to equation 2.25:

$$\begin{array}{ll}
 \text{ODE}_p: & \frac{L}{R} \cdot \frac{d}{dt} \underline{\hat{i}}_p + \underline{\hat{i}}_p = \frac{u_q}{R} \quad \text{with } u_q = \operatorname{Re} \left\{ \hat{\underline{U}}_q \cdot e^{j\omega t} \right\} \\
 \text{transformed:} & \Leftrightarrow \frac{L}{R} \cdot j\omega \underline{\hat{I}}_p + \underline{\hat{I}}_p = \frac{\hat{\underline{U}}_q}{R} \quad \text{with } \underline{\hat{U}}_q = \hat{U}_q \cdot e^{j\varphi_q}
 \end{array}$$

The approach offers a simple conversion from the complex amplitude pointer of the particular solution and is in principle equivalent to the derivation using complex voltage dividers as shown in equation 3.13 using the example of the RC element.

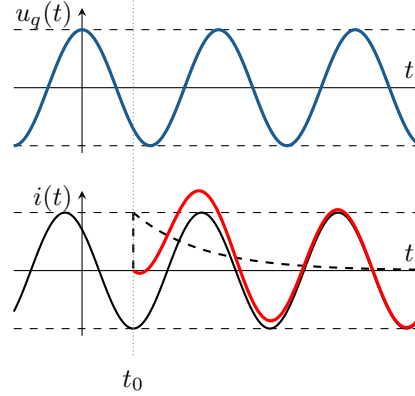
Example 3.5: RL element, switching operation with AC voltage source

Determination of the current i through the inductance L in series with the ohmic resistor R when the ideal AC voltage source $u_q = \hat{U}_q \cdot \cos(\omega t + \varphi_q)$ is switched on at the switching time $t = t_0$. No current flows prior to this time.



given: $i(t_0)$, u_q , R , L

wanted: $i(t)$, $u_L(t)$



1.) **Set up ODE:** from i for $t \geq t_0$, Solution from example. 3.3

$$\text{ODE:} \quad \frac{L}{R} \cdot \frac{d}{dt} i + i = \frac{u_q}{R} \quad \text{with} \quad u_q = \hat{U}_q \cdot \cos(\omega t + \varphi_q)$$

2.) **Homogeneous solution, gen.:** (fleeting), Solution from example 3.5 for switching time t_0

$$\text{hom. sol.:} \quad i_h = K \cdot e^{-\frac{t-t_0}{\tau}} \quad \text{with} \quad \tau = \frac{L}{R}$$

3.) **Particular solution:** (settled in) with $t \rightarrow \infty$, complete AC calculation

$$\text{transf.:} \quad \frac{L}{R} \cdot j\omega \hat{I}_p + \hat{I}_p = \frac{\hat{U}_q}{R} \quad \text{with} \quad \hat{U}_q = \hat{U}_q \cdot e^{j\varphi_q}$$

$$\text{Pointer:} \quad \hat{I}_p = \frac{\hat{U}_q}{R} \cdot \frac{1}{1 + j\omega \frac{L}{R}} \quad \text{with} \quad \hat{I}_p = \hat{I}_p \cdot e^{j\varphi_I}$$

$$\hat{I}_p = \frac{\hat{U}_q/R}{\sqrt{1 + (\omega \frac{L}{R})^2}} \quad \varphi_I = \varphi_q - \arctan\left(\omega \frac{L}{R}\right)$$

$$\text{part. sol.:} \quad i_p = \hat{I}_p \cdot \cos(\omega t + \varphi_I) \quad \text{with} \quad i_p = \text{Re} \left\{ \hat{I}_p \cdot e^{j\omega t} \right\}$$

4.) **Inhomogeneous solution, gen.:** (Superposition)

$$\text{gen. sol.:} \quad i = \underbrace{K \cdot e^{-\frac{t-t_0}{\tau}}}_{i_h} + \underbrace{\hat{I}_p \cdot \cos(\omega t + \varphi_I)}_{i_p}$$

5.) **Determine constants:** (Insert AB)

$$\text{Set initial cond.:} \quad i(t_0) = K \cdot e^0 + \hat{I}_p \cdot \cos(\omega t_0 + \varphi_I) \stackrel{!}{=} 0$$

$$\Rightarrow K = -\hat{I}_p \cdot \cos(\omega t_0 + \varphi_I)$$

$$\text{clear sol.:} \quad i(t) = -\hat{I}_p \cdot \cos(\omega t_0 + \varphi_I) \cdot e^{-\frac{t-t_0}{\tau}} + \hat{I}_p \cdot \cos(\omega t + \varphi_I)$$

Calculate voltage u_L across inductance L using $u_L = L \cdot \frac{d}{dt} i$.

$$\text{Voltage:} \quad u_L(t) = R \cdot \hat{I}_p \cdot \cos(\omega t_0 + \varphi_I) \cdot e^{-\frac{t-t_0}{\tau}} - \omega L \cdot \hat{I}_p \cdot \sin(\omega t + \varphi_I)$$

3.3 RLC series resonant circuits

Systems with two different energy storage devices (inductance L and capacitance C) are fundamentally capable of oscillation. For this reason, RLC circuits are also referred to as oscillating circuits or resonance circuits. The oscillatory capability can be determined and described using the eigenvalues. Another approach is to consider the (oscillating circuit) quality factor Q . [See Module 8]

Figure 3.6 shows the circuit diagram of an RLC element charging with a constant DC voltage U_q . The time curves of the capacitor voltage for the three possible cases are also shown. These are the **aperiodic case**, the **aperiodic limit case** and the **periodic case** which are described in more detail in Chapter 3.3.1.

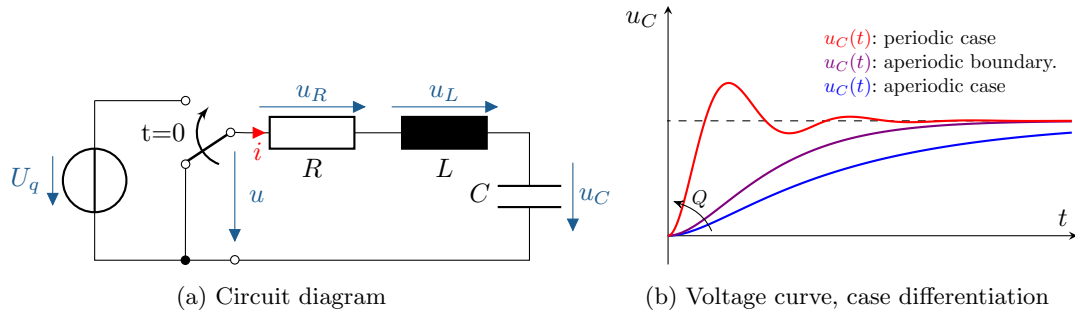


Figure 3.6: RLC element, charging process with constant DC voltage

The calculation of switching operations for RLC elements is analogous to the examples for RC and RL elements from the previous chapters.

1.) **Set up ODE:** for u_C for $t \geq 0$

$$\begin{aligned}
 u_L + u_R + u_C &= u \\
 L \cdot \frac{d}{dt} i + R \cdot i + u_C &= U_q \\
 LC \cdot \frac{d^2}{dt^2} u_C + RC \cdot \frac{d}{dt} u_C + u_C &= U_q
 \end{aligned} \tag{3.18}$$

Due to the two different energy storage devices, a second-order ODE results.

2.) **Homogeneous solution, gen.:** (fleeting) (without interference)

The form of the homogeneous solution depends on the type of eigenvalues.

$$\text{hom. ODE: } LC \cdot \frac{d^2}{dt^2} u_{C,h} + RC \cdot \frac{d}{dt} u_{C,h} + u_{C,h} = 0 \quad \text{with } u_{C,h} = K \cdot e^{\lambda t} \tag{3.19}$$

$$\text{char. Polynom: } \lambda^2 + \frac{R}{L} \cdot \lambda + \frac{1}{LC} = 0$$

$$\text{Eigenvalues: } \lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad \text{with } D = \left(\frac{R}{2L}\right)^2 - \frac{1}{LC} \tag{3.20}$$

Applying the p - q formula (midnight formula) yields either two different or two identical values for the **eigenvalues** (EW) $\lambda_{1/2}$, depending on the root term. The homogeneous solution therefore takes one of the following two forms, as described in Chapter 2.4.2:

$$\text{hom. sol.: } u_{C,h} = \begin{cases} K_1 \cdot e^{\lambda_1 t} + K_2 \cdot e^{\lambda_2 t} & \text{for } \lambda_1 \neq \lambda_2 \\ (K_1 + K_2 \cdot t) \cdot e^{\lambda t} & \text{for } \lambda_1 = \lambda_2 = \lambda \end{cases} \tag{3.21}$$

The cases can be distinguished using the so-called **discriminant** D .

3.3.1 Case differentiation for RLC series resonant circuits

The discriminant D is the term under the root in equation 3.20 for determining $\lambda_{1/2}$.

$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}} \quad (3.22)$$

Table 3.2 lists all three cases with the type of eigenvalues and relative statements regarding the damping of the oscillating circuit. The value ranges of the discriminant D and the oscillating circuit quality factor Q_S (see Module 8) are also given as distinguishing features.

Table 3.2: Case differentiation for the RLC resonant circuit

Case differentiation	Damping	Discrimi.	Quality factor	Eigenvalues $\lambda_{1/2}$
Aperiodic case	strong	$D > 0$	$Q_S < 0.5$	Two distinct real eigenvalues
Critical aperiodic case	critical	$D = 0$	$Q_S = 0.5$	One repeated real eigenvalue
Periodic case	weak	$D < 0$	$Q_S > 0.5$	Complex conjugate eigenvalues

A locus diagram of the eigenvalues is shown in the optional excursus in Chapter 3.4.2. The relationship between λ and Q_S is explained in more detail in the optional excursus in Chapter 3.4.3.

3.3.2 Aperiodic case in the RLC series resonant circuit

If the discriminant D is positive, the root term is real and always smaller than the absolute value of the term $\frac{R}{2L}$ before the root. As a result, both eigenvalues are different, real and negative.

$$\text{with } D > 0 \text{ applies} \quad \frac{R}{2L} > \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad \Rightarrow \quad 2 \text{ real eigenvalues}$$

2.) Homogeneous solution: (fleeting)

$$u_{C,h} = K_1 \cdot e^{\lambda_1 t} + K_2 \cdot e^{\lambda_2 t} \quad \text{with } \lambda_1 \neq \lambda_2, \lambda < 0 \quad (3.23)$$

The constants K_1 and K_2 must both be real so that the solution remains real-valued. The transient state thus corresponds to the addition of two exponential functions. The slower exponential function dominates, i.e. the one with the larger time constant (see excursus in chapter 3.4.2).

3.) Particular solution: (settled, identical for all three cases)

$$u_{C,p} = U_q$$

4.) General solution: (Superposition)

$$u_C(t) = K_1 \cdot e^{\lambda_1 t} + K_2 \cdot e^{\lambda_2 t} + U_q$$

5.) Determine constants: (as is known)

Be $u_C(0) = 0$ and $i(0) = 0 \Rightarrow \frac{d}{dt} u_C(0) = 0$ as initial conditions (IC) given, then it follows that

$$\begin{aligned} \text{1st IC:} \quad & u_C(0) = K_1 \cdot e^{\lambda_1 \cdot 0} + K_2 \cdot e^{\lambda_2 \cdot 0} + U_q \stackrel{!}{=} 0 \quad \Rightarrow \quad K_1 = -K_2 - U_q \\ \text{2nd IC:} \quad & \frac{d}{dt} u_C(0) = K_1 \cdot \lambda_1 \cdot e^{\lambda_1 \cdot 0} + K_2 \cdot \lambda_2 \cdot e^{\lambda_2 \cdot 0} + 0 \stackrel{!}{=} 0 \quad \Rightarrow \quad K_1 = -K_2 \frac{\lambda_2}{\lambda_1} \\ \Rightarrow \quad & K_1 = \frac{U_q}{\frac{\lambda_1}{\lambda_2} - 1} \quad \text{and} \quad K_2 = \frac{U_q}{\frac{\lambda_2}{\lambda_1} - 1} \end{aligned}$$

3.3.3 Aperiodic boundary case in the RLC series resonant circuit

If the discriminant D is exactly zero, both eigenvalues are identical, real and negative. The double zero point results in a special form of the homogeneous solution by superimposing the two approaches ($K_1 \cdot e^{\lambda_1 t}$ and $K_2 \cdot t \cdot e^{\lambda_2 t}$).

$$\text{for } D = 0 \text{ applies } \lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{0} = \lambda \quad \Rightarrow \quad 1 \text{ real Eigenvalue}$$

2.) Homogeneous solution: (fleeting)

$$u_{C,h} = (K_1 + K_2 \cdot t) \cdot e^{\lambda t} \quad \text{with } \lambda_1 = \lambda_2 < 0 \quad (3.24)$$

3.) Particular solution: (settled in)

$$u_{C,p} = U_q$$

4.) General solution: (settled in)

$$u_C(t) = (K_1 + K_2 \cdot t) \cdot e^{\lambda t} + U_q$$

5.) Determine constants: (as is known)

Let $u_C(0) = 0$ and $i(0) = 0 \Rightarrow \frac{d}{dt} u_C(0) = 0$ be given as initial conditions (IC), then the following applies:

$$\begin{aligned} \text{1st IC: } u_C(0) &= (K_1 + K_2 \cdot 0) \cdot e^{\lambda \cdot 0} + U_q \stackrel{!}{=} 0 & \Rightarrow & K_1 = -U_q \\ \text{2nd IC: } \frac{d}{dt} u_C(0) &= (K_1 + K_2 \cdot 0) \cdot \lambda \cdot e^{\lambda \cdot 0} + K_2 \cdot e^{\lambda \cdot 0} \stackrel{!}{=} 0 & \Rightarrow & K_2 = -K_1 \cdot \lambda \\ \Rightarrow & K_1 = -U_q \quad \text{and} \quad K_2 = +U_q \cdot \lambda \end{aligned}$$

3.3.4 Periodic case for the RLC series resonant circuit

If the discriminant is negative, there are two complex conjugate eigenvalues with negative real parts. Transformation with $\sqrt{-1} = j \cdot \sqrt{1}$ and derivation for $u_{C,h}$ in chapter 3.4.1:

$$\text{for } D < 0 \text{ applies } \lambda_{1/2} = -\underbrace{\frac{R}{2L}}_{\delta} \pm j \cdot \underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{\omega_d} \quad \Rightarrow \quad 2 \text{ compl. EW} \quad (3.25)$$

2.) Homogeneous solution: (fleeting)

$$u_{C,h} = e^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) \quad \text{with } \lambda_{1/2} = -\delta \pm j \omega_d \quad (3.26)$$

The homogeneous solution corresponds to a decaying oscillation with the **decay constant** δ (negative real part of the EW) and the **damped natural frequency** ω_d (positive imaginary part of the EW). A derivation for Eq. 3.26 from Eq. 3.28 is given in Chapter 3.4.1.

3.) Particular solution: (settled in)

$$u_{C,p} = U_q$$

4.) General solution: (Superposition)

$$u_C(t) = e^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) + U_q$$

5.) Determine constants: (as is known)

1st IC: $u_C(0) = e^{-\delta \cdot 0} \cdot C_0 \cdot \sin(\omega_d \cdot 0 + \varphi_0) + U_q \stackrel{!}{=} 0$

2nd IC: $\frac{d}{dt}u_C(0) = -\delta \cdot e^{-\delta \cdot 0} \cdot C_0 \cdot \sin(\varphi_0) + e^{-\delta \cdot 0} \cdot C_0 \cdot \omega_d \cdot \cos(\varphi_0) \stackrel{!}{=} 0$

$$\Rightarrow C_0 = -\frac{U_q}{\sin(\varphi_0)} \quad \text{and} \quad \varphi_0 = \arctan\left(\frac{\omega_d}{\delta}\right)$$

Figure 3.7 shows the time curve of the voltage $u_C(t)$ for $t = 0$ to $t = 5 \cdot \frac{1}{\delta}$. In the selected scaling, ω_0 is four times the decay constant δ (quality factor $Q_S = 2$).

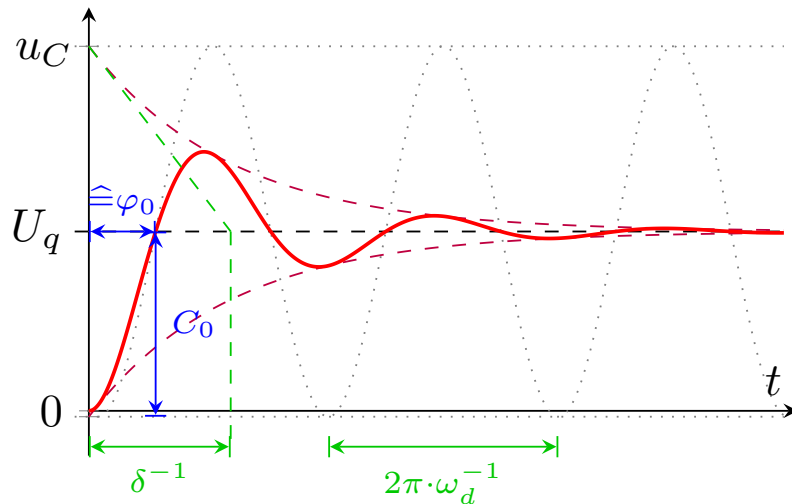


Figure 3.7: Voltage $u_C(t)$ in periodic cases with associated constants: C_0 , φ_0 , δ , ω_d

3.4 Digressions on series resonant circuits

This chapter contains supplementary digressions on the behaviour of oscillating circuits, which serve to deepen understanding of the topic and build bridges to other subject areas. The content of the digressions is not essential for understanding the rest of the module and can be skipped if necessary.

3.4.1 Derivation of the homogeneous solution in the periodic case

From the general homogeneous solution in the periodic case (Eq. 3.21), it follows with $\lambda_{1/2}$ (Eq. 3.25):

$$u_{C,h} = e^{-\delta t} \cdot (K_1 \cdot e^{+j\omega_d t} + K_2 \cdot e^{-j\omega_d t}) \quad (3.27)$$

K_1 and K_2 must be complex conjugates so that $u_{C,h}$ is real! [2, cf. p. 379]

$$\text{with} \quad K_1 = a + jb \quad \text{and} \quad K_2 = a - jb \quad \text{with} \quad a, b \in \mathbb{R}$$

$$\begin{aligned} u_{C,h} &= e^{-\delta t} \cdot [(a + jb) \cdot e^{+j\omega_d t} + (a - jb) \cdot e^{-j\omega_d t}] \\ u_{C,h} &= e^{-\delta t} \cdot \left[a \cdot \underbrace{(e^{+j\omega_d t} + e^{-j\omega_d t})}_{2 \cos(\omega_d t)} + jb \cdot \underbrace{(e^{+j\omega_d t} - e^{-j\omega_d t})}_{2j \sin(\omega_d t)} \right] \end{aligned}$$

$$\begin{aligned} \text{with} \quad e^{+jx} + e^{-jx} &= \cos(x) + j\sin(x) + \cos(x) - j\sin(x) = 2 \cos(x) \\ e^{+jx} - e^{-jx} &= \cancel{\cos(x)} + j\sin(x) - \cancel{\cos(x)} + j\sin(x) = 2j \sin(x) \end{aligned}$$

$$u_{C,h} = e^{-\delta t} \cdot \left[\underbrace{2a}_{C_1} \cdot \cos(\omega_d t) - \underbrace{2b}_{C_2} \cdot \sin(\omega_d t) \right] \quad (3.28)$$

In periodic cases, the homogeneous solution corresponds to a decaying ($e^{-\delta t}$) superposition of the oscillations z_1 and $-z_2$ with the same angular frequency ω_d and a phase shift of $\pi/2$ between them.

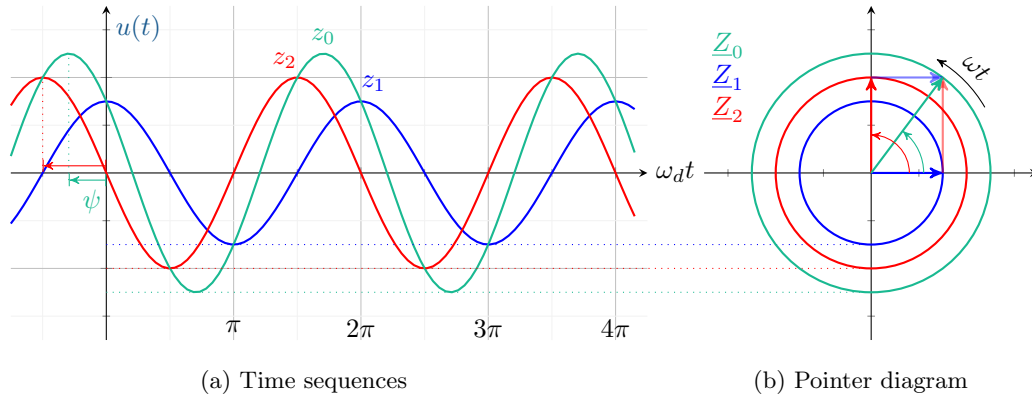


Figure 3.8: Superposition of oscillations of the same frequency, time course and phasor diagram

Figure 3.8 shows the superposition in the form of added time courses (a) and in the form of added complex amplitude pointers in the pointer diagram (b). Mathematically expressed:

<i>Time domain</i>	$\Leftrightarrow$	<i>Polarform</i>	$\Leftrightarrow$	<i>Cartesian</i>
$z_1(t) = +C_1 \cdot \cos(\omega_d t)$	$\Leftrightarrow$	$\underline{z}_1 = C_1 \cdot e^{j0}$	$\Leftrightarrow$	$\underline{z}_1 = C_1$
$z_2(t) = -C_2 \cdot \sin(\omega_d t)$	$\Leftrightarrow$	$\underline{z}_2 = C_2 \cdot e^{-j\frac{\pi}{2}}$	$\Leftrightarrow$	$\underline{z}_2 = -j \cdot C_2$
$z_0(t) = z_1(t) + z_2(t)$	$\Leftrightarrow$	$\underline{z}_0 = \underline{z}_1 + \underline{z}_2$	$\Leftrightarrow$	$\underline{z}_0 = C_1 - j \cdot C_2$
$= C_0 \cdot \cos(\omega_d t + \psi)$	$\Leftrightarrow$	$= C_0 \cdot e^{j\psi}$		

with

$C_0 = \sqrt{C_1^2 + C_2^2}$

and

$\psi = \arctan\left(-\frac{C_2}{C_1}\right), \quad \varphi_0 = \psi + \frac{\pi}{2}$

(3.29)

$$z_0(t) = C_0 \cdot \sin(\omega_d t + \varphi_0)$$

The homogeneous solution in the periodic case from Eq. 3.28 can be described according to Eq. 3.29 as a decaying, phase-shifted sine curve, as shown below (cf. Eq. 3.26):

$$u_{C,h} = e^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) \quad \text{with} \quad \lambda_{1/2} = -\delta \pm j \cdot \omega_d$$

3.4.2 Digression: Case differentiation for RLC series resonant circuits, zero points

The zero points of the characteristic polynomial, i.e. the eigenvalues $\lambda_{1/2}$ of the RLC series oscillating circuit, depend on the discriminant D .

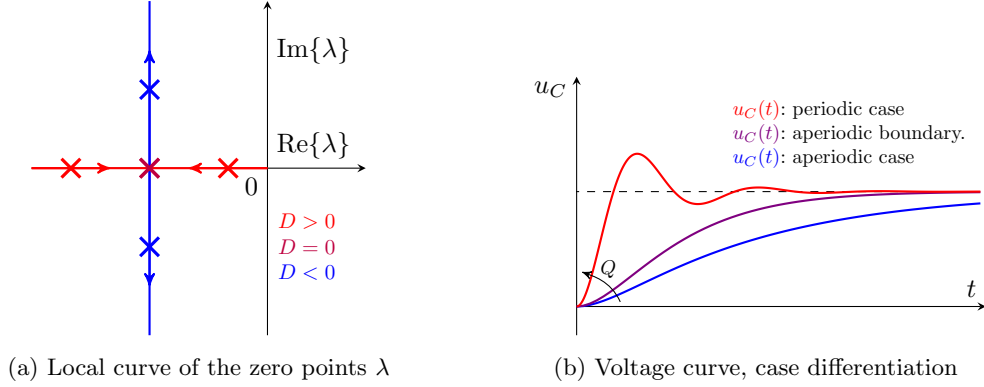


Figure 3.9: Local curve of eigenvalues and stress curves in the RLC series resonant circuit

Figure 3.9a shows the location of the zero points in the complex number space as a locus curve as a function of D for $\delta = \frac{R}{2L} = \text{const.}$. The course of the NST corresponds to an increase in the undamped natural frequency $\omega_0 = \frac{1}{\sqrt{LC}}$. [See Eq. 3.35]

In the aperiodic case ($D > 0$), the two negative real zeros for $\delta = \text{const.}$ and increasing ω_0 move towards each other. The zero points closer to the imaginary axis dominates the transient behaviour of the system (slower decay process). In the aperiodic limit case ($D = 0$), the zeros merge into a double zero at $\lambda_{1/2} = -\delta$. In the periodic limit case ($D < 0$), the conjugate complex zeros for $\delta = \text{const.}$ and increasing ω_0 move parallel to the imaginary axis and away from the real axis.

Comparison: Pole locations of the transfer function (system theory)

The position of the eigenvalues in the locus diagram allows conclusions to be drawn about the transient behaviour of the RLC series resonant circuit. This procedure is common in the stability analysis of systems in control engineering and originates from system theory.

For vanishing initial conditions ($\frac{d^i}{dt^i} u_C(0) = 0 \forall i$), the differential equation from Eq. 3.18 can be described directly in the image domain using the Laplace transform. By rearranging the transformed output variable $U_C(s)$ according to the input variable $U(s)$, the transfer function $G(s)$ of the RLC series resonant circuit is obtained:

$$u(t) = LC \cdot \frac{d^2}{dt^2} u_C(t) + RC \cdot \frac{d}{dt} u_C(t) + u_C(t) \quad \Bigg| \mathcal{L} \quad (3.30)$$

$$U(s) = LC \cdot s^2 \cdot U_C(s) + RC \cdot s \cdot U_C(s) + U_C(s) \quad (3.31)$$

$$G(s) = \frac{U_C(s)}{U(s)} = \frac{\text{Output variable}(s)}{\text{Input variable}(s)} \quad (3.32)$$

$$= \frac{1}{LC \cdot s^2 + RC \cdot s + 1} \quad (3.33)$$

The poles of the transfer function $G(s)$ correspond to the eigenvalues $\lambda_{1/2}$.

3.4.3 Decay constant, natural and resonance circuit frequency, quality factor

In the periodic case, the (damped) natural frequency ω_d and the decay constant δ are derived from the eigenvalues $\lambda_{1/2}$. These quantities are closely related to the **quality factor** Q_S and the **resonant frequency** ω_0 of an oscillating circuit (resonant circuit). Both quantities are used as parameters for oscillating circuits and are discussed in more detail in Module 8 (Frequency-variable circuits).

The relationship between ω_d , δ , ω_0 and Q_S will be explained using the RLC series oscillating circuit from Figure 3.6a. According to Equation 3.25, the eigenvalues $\lambda_{1/2}$ are given by:

$$\lambda_{1/2} = - \underbrace{\frac{R}{2L}}_{\delta} \pm j \cdot \underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{\omega_d} \quad (3.34)$$

The negative real part $-\delta$ determines the decay rate and the imaginary part ω_d determines the angular frequency of the linearly damped harmonic oscillation. The natural angular frequency ω_d can also be expressed as a function of ω_0 and δ using Eq. 3.39.

$$\omega_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} = \sqrt{\omega_0^2 - \delta^2} \quad (3.35)$$

This has the advantage that the undamped natural frequency ω_0 and the decay constant δ can be read directly from the coefficients of the differential equation. The following form is suitable for this purpose:

$$\ddot{y} + 2\delta \cdot \dot{y} + \omega_0^2 \cdot y = b \quad (3.36)$$

for ordinary, homogeneous, linear second-order differential equations. The differential equation describes the behaviour of linear damped second-order systems (with two energy stores). The undamped natural frequency corresponds to the resonance frequency ω_0 .

Resonance occurs when an oscillating circuit is excited at its **resonance frequency** f_0 . In the case of resonance, the reactive component of the total impedance $\underline{Z}(\omega)$ disappears (resonance condition). [See Module 8]

With the total impedance:

$$\underline{Z}(\omega) = R + j \left(\omega L - \frac{1}{\omega C} \right) \quad (3.37)$$

and the resonance condition:

$$\text{Im}\{\underline{Z}(\omega_0)\} = \omega_0 L - \frac{1}{\omega_0 C} \stackrel{!}{=} 0 \quad (3.38)$$

follows for the resonance circuit frequency:

$$\omega_0 = \sqrt{\frac{1}{LC}} \quad (3.39)$$

The quality factor Q_S , which measures the oscillation capability of an oscillating circuit, can also be derived from ω_0 and δ . It is defined as the ratio of the reactive power oscillating within the circuit to the power loss within the circuit in the case of resonance. In the case of an RLC series oscillating circuit, this corresponds to the ratio of the reactive resistance of the inductance $X_{L,0}$ at the resonance frequency to the series resistance R (without derivation). This results in the following relationship:

$$Q_s = \frac{X_{L,0}}{R} = \frac{\omega_0 L}{R} = \frac{\omega_0}{2\delta} \quad (3.40)$$

The quality is proportional to the ratio of the undamped natural frequency to the damping constant.

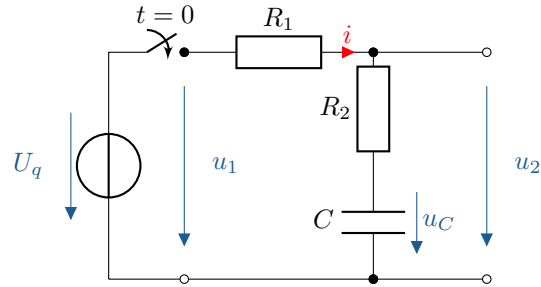
A Exercise tasks

A.1 Switch-on behaviour of an RC transmission element

The transfer element (two-stage) shown on the right is given. The transient behaviour of the voltage u_2 at the output is examined when an ideal DC voltage source is connected. U_q .

Before the switching point $t = 0$, the circuit is in a steady state with $u_2(t < 0) = 0$.

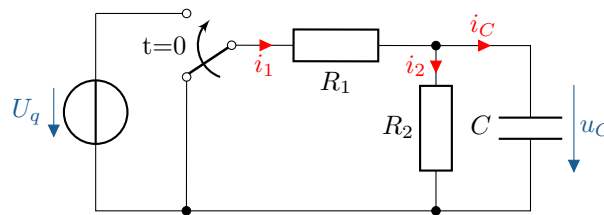
- Set up the ODE for u_C for $t \geq 0$.
- Determine the function $u_C(t)$ for $t \geq 0$, including all constants.
- Determine the function u_2 for $t \geq 0$ from the solution b) for $u_C(t)$. Simplify as much as possible.



- Sketch the qualitative time curves of the voltages u_2 , u_{R_2} and u_C from the steady state for $t < 0$ until the represented variables have stabilised ($t = 5\tau$). Identify the time constants, initial and final values of the curves.

A.2 Switch-on behaviour (DC) of a lossy capacitor

The switch-on behaviour of the circuit shown is to be investigated.

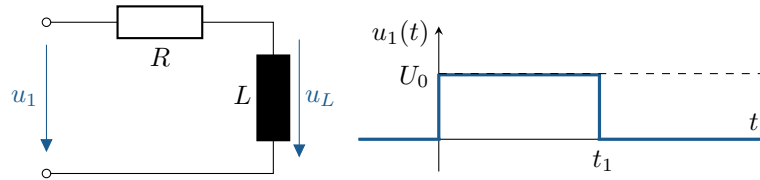


At time $t = 0$, the switch is flipped so that the DC voltage U_q is applied to the circuit. The capacitance is completely discharged before the switching point.

- Determine the equation for the voltage across the capacitor $u_C(t)$ for $t \geq 0$.
- Plot the time course of the voltages $u_C(t)$ and $u_{R_1}(t)$ as well as the current $i_C(t)$. Mark characteristic points.
- After the capacitor C is fully charged to U_0 , the switch is returned to its initial position. How long does it take for the capacitor C to discharge to 10 % of the voltage U_0 ?
- The capacitor is discharged as in c), but with the switch open (no-load). How long does the discharge process take (from U_0 to $10\% \cdot U_0$)? Compare the duration with that from c).

A.3 Switching behaviour of an RL series connection

The switching behaviour of the circuit shown in the left-hand figure is to be investigated. The voltage curve $u_1(t)$ is shown in the right-hand figure.



Before time $t = 0$, the voltage $u_1 = 0$, so that at time $t = 0$ no energy is stored in the coil. Furthermore, $t_1 \gg 5\tau$.

- Consider the charging process ($0 < t < t_1$). Given are $\tau = 1 \text{ ms}$ and $R = 100 \Omega$. What is the inductance L ?
- Continue to consider the charging process ($0 < t < t_1$). Now the voltage is given as $U_0 = 1 \text{ kV}$. What is the current through the inductor i_L at time $t = 3\tau$?
- Continue to consider the charging process. Now: $R = 5 \Omega$, $L = 100 \text{ mH}$, and $U_0 = 2.8 \text{ kV}$.
 - After what time t_0 is the transient response completed?
 - What are the inductor current i_L and inductor voltage u_L after $t = 50 \text{ ms}$?
- Now consider the discharging process ($t > t_1$). The values from the previous charging part apply. What are the inductor current i_L and inductor voltage u_L at $t = t_1 + 50 \text{ ms}$?
- Plot the time courses of the inductor voltage u_L , the current i_L , and the resistor voltage u_R . Mark the time constant τ and other characteristic points. Assume $t_1 = 120 \text{ ms}$.

A.4 Switch-on behaviour of an RL series connection

The switch-on behavior of a series circuit consisting of a resistive resistor R and an inductance $L = 100 \text{ mH}$ is to be examined. At time $t = 0$, a voltage $u = U_0 = 35 \text{ V}$ is applied to the series circuit. For $t \leq 0$, $u = 0$. The inductor voltage at time $t_1 = 3 \text{ ms}$ is known and amounts to $u_L(t_1) = 26 \text{ V}$.

Determine the time constant τ , the resistance R , and the voltage $u_R(t = 5 \text{ ms})$.

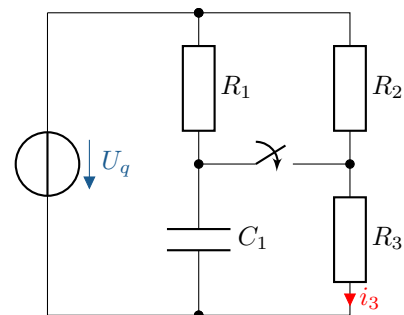
A.5 ET3 Ü8 Switching behaviour of an RC network

The switch in the circuit shown on the right is closed at time $t = 0$.

$$\begin{aligned} R_1 &= 1,2 \text{ k}\Omega & R_2 &= 2 \text{ k}\Omega \\ C_1 &= 1 \mu\text{F} & R_3 &= 500 \Omega \\ U_q &= 250 \text{ V} \end{aligned}$$

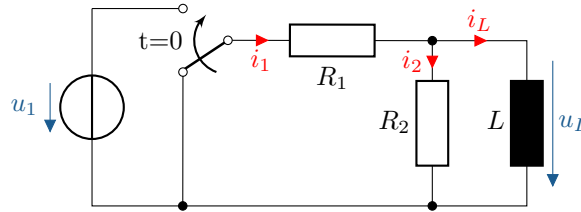
Determine the time course of the current i_3 through the resistor R_3 and plot it as a function of time (line graph) for $-\tau < t < 5\tau$.

The network is in a steady state at time $-\tau$. Here, τ is the time constant of the network.



A.6 Switching process for alternating voltage

At time $t = t_0$, the alternating voltage $u_1(t)$ is applied to the entire circuit consisting of R_1 in series with the parallel connection of R_2 and C . For $t < t_0$, the circuit is in a steady state. The source voltage is given by $u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$.



- What current $i_{L,e}(t)$ occurs in the steady state at the inductance? Determine the amplitude and zero phase angle of the current.
- Set up the ODE of $i_L(t)$ for $t \geq t_0$. What is the time constant for the transient state? Note (ODE): $\tau \cdot \frac{d}{dt} i_L + i_L = \text{disturbance}$
- When must switching occur in order to minimise the transient state of $i_L(t)$? At which possible switching times does this become maximum? Give the answer using the phase shift φ between $u_1(t)$ and $i_L(t)$.
- At the switching instant t_0 , let the steady-state value of $i_{L,e}(t)$ be at its negative peak. The time constant τ equals one period $T = \frac{2\pi}{\omega}$. Determine the current maximum during the transient response.

B Solutions to the exercises

B.1 Switch-on behaviour of an RC transmission element

- a) ODE for $u_C(t \geq 0)$:

$$R_1 \cdot i + R_2 \cdot i + u_C = U_q$$

$$\underbrace{(R_1 + R_2) \cdot C}_{=\tau} \cdot \frac{d}{dt} u_C + u_C = U_q$$

- b) Time course u_C for $t \geq 0$:

$$u_{C,f} = u_{C,h} = K \cdot e^{-\frac{t}{\tau}} \quad \text{with } \tau = (R_1 + R_2) \cdot C$$

$$u_{C,e} = u_{C,p} = U_q$$

$$u_C = u_{C,f} + u_{C,e} = K \cdot e^{-\frac{t}{\tau}} + U_q$$

$$u_C(0) = K \cdot \cancel{e^0} + U_q \stackrel{!}{=} 0 \quad \implies K = -U_q$$

$$u_C(t) = \left(1 - e^{-\frac{t}{\tau}}\right) \cdot U_q$$

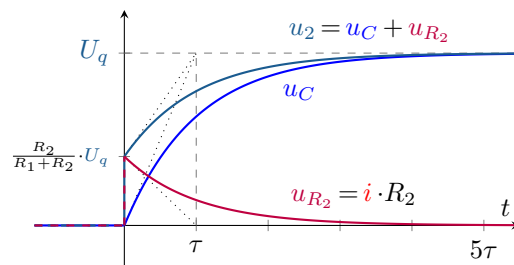
c) Voltage $u_2(t)$ from $u_C(t)$ for $t \geq 0$:

$$\begin{aligned}
 u_2 &= i \cdot R_2 + u_C \\
 &= C \cdot R_2 \cdot \frac{d}{dt} u_C + u_C \\
 &= C \cdot R_2 \cdot \frac{d}{dt} \left(1 - e^{-\frac{t}{\tau}} \right) \cdot U_q + \left(1 - e^{-\frac{t}{\tau}} \right) \cdot U_q \\
 &= \frac{C \cdot R_2}{\tau} \cdot e^{-\frac{t}{\tau}} \cdot U_q + \left(1 - e^{-\frac{t}{\tau}} \right) \cdot U_q \\
 &= \frac{R_2}{R_1 + R_2} \cdot e^{-\frac{t}{\tau}} \cdot U_q + \left(1 - e^{-\frac{t}{\tau}} \right) \cdot U_q \\
 &= \left(1 + \frac{R_2}{R_1 + R_2} \cdot e^{-\frac{t}{\tau}} - e^{-\frac{t}{\tau}} \right) \cdot U_q \\
 &= \left(1 - \frac{R_1}{R_1 + R_2} \cdot e^{-\frac{t}{\tau}} \right) \cdot U_q
 \end{aligned}$$

d) Sketch for u_C , u_2 and u_{R_2} .

$$u_{R_2}(0^-) = u_2(0^-) = 0$$

$$u_{R_2}(0^+) = u_2(0^+) = \frac{R_2}{R_1 + R_2} \cdot U_q$$



B.2 Switch-on behaviour (DC) of a lossy capacitor

a) Initial state ($t \geq 0$): U_q applied to the circuit: R_1 in series with a parallel combination of R_2 and C .

1. Set up the differential equation for u_C ($t \geq 0$)

$$\begin{aligned}
 u_{R1} + u_C &= U_q & u_{R1} &= R_1 \cdot i_{R1} \\
 R_1 \cdot i_{R1} + u_C &= U_q & i_{R1} &= i_{R2} + i_C \\
 R_1 \cdot (i_C + i_{R2}) + u_C &= U_q & i_C &= C \cdot \frac{d}{dt} u_C & i_{R2} &= \frac{u_C}{R_2} \\
 C \cdot R_1 \cdot \frac{d}{dt} u_C + \frac{R_1}{R_2} \cdot u_C + u_C &= U_q & & \left| \cdot R_2 \right. \\
 C \cdot R_1 \cdot R_2 \cdot \frac{d}{dt} u_C + (R_1 + R_2) \cdot u_C &= U_q \cdot R_2 & & \left| : (R_1 + R_2) \right. \\
 \underbrace{C \cdot \frac{R_1 \cdot R_2}{R_1 + R_2}}_{\tau} \cdot \frac{d}{dt} u_C + u_C &= U_q \cdot \frac{R_2}{R_1 + R_2}
 \end{aligned}$$

2. Homogeneous solution and 3. Particulate solution ($t \rightarrow \infty$)

$$\begin{aligned}
 u_{C,h} &= K \cdot e^{\lambda t} = K \cdot e^{-\frac{t}{\tau}} & \tau &= C \cdot \frac{R_1 \cdot R_2}{R_1 + R_2} \\
 u_{C,p} &= U_q \cdot \frac{R_2}{R_1 + R_2} & C &\text{entspricht Leerlauf}
 \end{aligned}$$

4. Determine superposition and 5th constant K

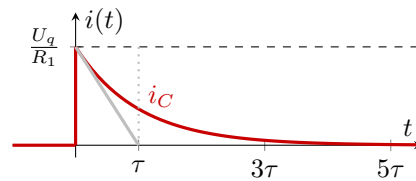
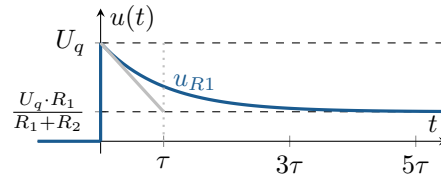
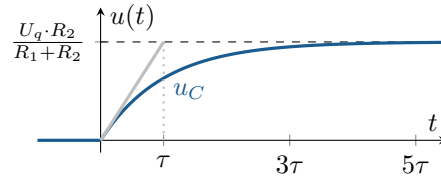
$$\begin{aligned}
 u_C(t) &= u_{C,h} + u_{C,p} = K \cdot e^{-\frac{t}{\tau}} + U_q \cdot \frac{R_2}{R_1 + R_2} \\
 u_C(0) &= K \cdot \cancel{e^0} + U_q \cdot \frac{R_2}{R_1 + R_2} \stackrel{!}{=} 0 & \Rightarrow K &= -U_q \cdot \frac{R_2}{R_1 + R_2} \\
 u_C(t) &= U_q \cdot \frac{R_2}{R_1 + R_2} \cdot \left(1 - e^{-\frac{t}{\tau}}\right)
 \end{aligned}$$

b) Sketch $u_C(t)$, u_{R_1} and i_C with $u_C(t)$ from a):

Time sequences:

$$\begin{aligned} u_{R_1}(t) &= U_q - u_C(t) \\ &= U_q - U_q \cdot \frac{R_2}{R_1 + R_2} \cdot \left(1 - e^{-\frac{t}{\tau}}\right) \\ &= U_q \cdot \frac{R_1}{R_1 + R_2} + U_q \cdot \frac{R_2}{R_1 + R_2} \cdot e^{-\frac{t}{\tau}} \end{aligned}$$

$$\begin{aligned} i_C(t) &= C \cdot \frac{d}{dt} u_C(t) \\ &= C \cdot U_q \cdot \frac{R_2}{R_1 + R_2} \cdot \frac{1}{\tau} \cdot e^{-\frac{t}{\tau}} \\ &= U_q \cdot \frac{\cancel{C} \cdot \cancel{R_2}}{\cancel{R_1 + R_2}} \cdot \frac{\cancel{R_1 + R_2}}{\cancel{C} \cdot \cancel{R_1} \cdot \cancel{R_2}} \cdot e^{-\frac{t}{\tau}} \\ &= \frac{U_q}{R_1} \cdot e^{-\frac{t}{\tau}} \end{aligned}$$



c) Discharging of C through the parallel combination of R_1 and R_2 for $t' \geq 0$ until $U_0 \cdot 10\%$
Discharge time constant: $\tau_1 = \tau = C \cdot \frac{R_1 \cdot R_2}{R_1 + R_2}$:

$$\begin{aligned} u_C(t') &= U_0 \cdot e^{-\frac{t'}{\tau}} \stackrel{!}{=} U_0 \cdot 10\% \\ e^{-\frac{t'}{\tau}} &= \frac{1}{10} \\ -\frac{t'}{\tau} &= \ln\left(\frac{1}{10}\right) = -\ln(10) \\ t' &= \ln(10) \cdot \tau \approx 2,3025 \cdot \tau \end{aligned}$$

d) Discharging of C through R_2 , when the switch is open (no-load) for $t' \geq 0$ until $U_0 \cdot 10\%$.
Solution as in task c), but with discharge time constant $\tau_2 = R_2 \cdot C$.
The capacitor discharges more slowly than in task c) ($\tau_2 > \tau$), since $R_2 > R_1 || R_2$.

B.3 Switching behaviour of an RL series connection

$$\text{ODE (general):} \quad \frac{L}{R} \cdot \frac{d}{dt} i = \frac{U_1}{R_1} \quad (\text{Switch on})$$

$$\text{Solution approach:} \quad i = \frac{U_0}{R} \cdot (1 - e^{-\frac{t}{\tau}}) \quad \text{with} \quad \tau = \frac{L}{R}$$

a) Calculate inductance.

$$\tau = \frac{L}{R} \quad L = \tau \cdot R = 1 \text{ ms} \cdot 100 \Omega = 0,1 \text{ H}$$

b) Charging process, coil current i_L at time $t = 3\tau$.

$$\begin{aligned}
 i(t = 3 \cdot \tau) &= \frac{U_0}{R} (1 - e^{-\frac{t}{\tau}}) \\
 &= \frac{1 \text{ kV}}{100 \Omega} \cdot (1 - e^{-3}) = 10 \text{ A} \cdot 0,95 \\
 &= 9,5 \text{ A}
 \end{aligned}$$

c) Time until completion of the settling process, coil current and voltage after 50 ms.

$$\tau = \frac{L}{R} = \frac{0,1 \text{ H}}{5 \Omega} = 0,02 \text{ s} = 20 \text{ ms}$$

$$\begin{aligned}
 i(t = 50 \text{ ms}) &= \frac{U_0}{R} \cdot (1 - e^{-\frac{t}{\tau}}) \\
 &= \frac{2,8 \text{ kV}}{5 \Omega} \cdot (1 - e^{-\frac{50 \text{ ms}}{20 \text{ ms}}}) \\
 &= 0,56 \text{ kA} \cdot (0,918) = 0,514 \text{ kA}
 \end{aligned}$$

$$\begin{aligned}
 u_L &= L \cdot \frac{d}{dt} i_L = \frac{L}{R} \cdot U_0 \cdot \left(-\frac{1}{\tau}\right) \cdot (-e^{-\frac{t}{\tau}}) \\
 &= U_0 \cdot e^{-\frac{t}{\tau}}
 \end{aligned}$$

$$\begin{aligned}
 u_L(t = 50 \text{ ms}) &= 2,8 \text{ kV} \cdot e^{-\frac{50 \text{ ms}}{20 \text{ ms}}} = 229,8 \text{ V} \\
 t_0 &= 5 \cdot \tau = 5 \cdot 20 \text{ ms} = 100 \text{ ms}
 \end{aligned}$$

d) Discharge process, coil current and voltage after $t = t_1 + 50 \text{ ms}$.

$$\text{ODE (Unloading process):} \quad \frac{L}{R} \cdot \frac{d}{dt} i + i = 0$$

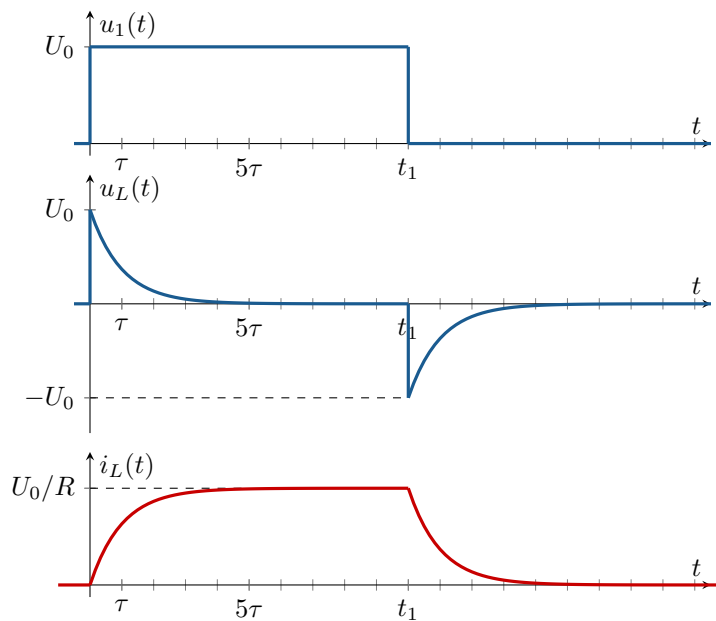
$$\text{Approach to a solution:} \quad i = \frac{U_0}{R} \cdot e^{-\frac{t}{\tau}} \quad \text{mit} \quad \tau = \frac{L}{R}$$

$$\begin{aligned}
 u_L &= L \cdot \frac{d}{dt} i = \frac{L}{R} \cdot u_0 \cdot \left(-\frac{1}{\tau}\right) \cdot e^{-\frac{t}{\tau}} \\
 &= -U_0 \cdot e^{-\frac{t}{\tau}}
 \end{aligned}$$

$$i(50 \text{ ms}) = \frac{2,8 \text{ kV}}{5 \Omega} \cdot e^{-\frac{50 \text{ ms}}{20 \text{ ms}}} = 46 \text{ A}$$

$$u_L(50 \text{ ms}) = -2,8 \text{ kV} \cdot e^{-\frac{50 \text{ ms}}{20 \text{ ms}}} = -230 \text{ V}$$

e) Temporal progression of u_L, i_L, u_R .



B.4 Switch-on behaviour of an RL series connection

$$\begin{aligned} \text{ODE:} \quad & \frac{L}{R} \cdot \frac{d}{dt} i + i = \frac{U_0}{R} \\ \text{Solution:} \quad & i = \frac{U_0}{R} \cdot (1 - e^{-\frac{t}{\tau}}) \quad \text{with} \quad \tau = \frac{L}{R} \end{aligned}$$

$$\begin{aligned} u_L &= L \cdot \frac{d}{dt} i = U_0 \cdot e^{-\frac{t}{\tau}} \\ u_R &= u - u_L = U_0 \cdot (1 - e^{-\frac{t}{\tau}}) \\ i(t = 3ms) &= \frac{U_0}{R} \cdot (1 - e^{-\frac{t}{\tau}}) \\ u_L(t = 3ms) &= U_0 \cdot e^{-\frac{t}{\tau}} = 26V \\ \frac{u_L(t = 3ms)}{U_0} &= e^{-\frac{t}{\tau}} \\ -\frac{t}{\tau} &= \ln\left(\frac{u_L(t = 3ms)}{U_0}\right) \\ \tau &= -\frac{t}{\ln\left(\frac{u_L(t = 3ms)}{U_0}\right)} = -\frac{3ms}{\ln\left(\frac{26V}{35V}\right)} = 10,1ms \\ R &= \frac{L}{\tau} = \frac{100mH}{10,1ms} = 9,9\Omega \\ u_R(t = 5ms) &= U_0 \cdot (1 - e^{-\frac{5ms}{10,1ms}}) \\ &= 35V \cdot 0,39 = 13,65V \end{aligned}$$

B.5 ET3 Ü8 Switching behaviour of an RC network

Set up the ODE for u_C and determine the current i_3 via $i_3 = \frac{u_C}{R_3}$.
Combine R_1 and R_2 as a parallel connection R_{12} after the switching instant ($t > 0$).

1st ODE for u_C ($t > 0$) with $i_3 = \frac{u_C}{R_3}$:

$$\begin{aligned}
 U_q &= u_{12} + u_C && \text{with } u_{12} \text{ via parallel connection of } R_1 \text{ and } R_2 \\
 &= R_{12} \cdot i_{12} + u_C && \text{with } R_{12} = R_1 || R_2, \quad i_{12} = i_C + i_3 \\
 &= R_{12} \cdot (i_C + i_3) + u_C && \text{with } i_C = C \cdot \frac{d}{dt} u_C, \quad i_3 = \frac{u_C}{R_3} \\
 &= R_{12} \cdot \left(C \cdot \frac{d}{dt} u_C + \frac{u_C}{R_3} \right) + u_C \\
 &= C \cdot R_{12} \cdot \frac{d}{dt} u_C + \left(\frac{R_{12}}{R_3} + 1 \right) \cdot u_C && \left| \cdot R_3 \right. \\
 U_q \cdot R_3 &= C \cdot R_{12} \cdot R_3 \cdot \frac{d}{dt} u_C + (R_{12} + R_3) \cdot u_C && \left| : (R_{12} + R_3) \right. \\
 U_q \cdot \frac{R_3}{R_{12} + R_3} &= \underbrace{C \cdot \frac{R_{12} \cdot R_3}{R_{12} + R_3}}_{\tau} \cdot \frac{d}{dt} u_C + u_C && \Rightarrow \tau = C \cdot \frac{R_{12} \cdot R_3}{R_{12} + R_3}
 \end{aligned}$$

3. Homogeneous solution (fleeting) and **2. Particular solution** ($t \rightarrow \infty$, settled in):

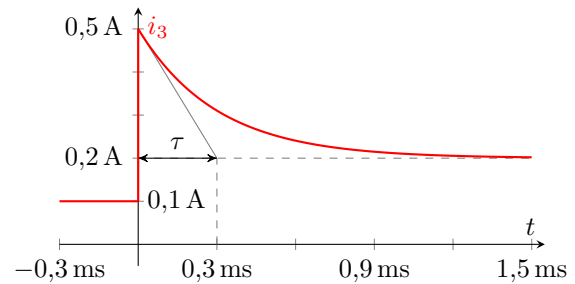
$$\begin{aligned}
 u_{C,h} &= K \cdot e^{-\frac{t}{\tau}} && \text{with } \tau = C \cdot \frac{R_{12} \cdot R_3}{R_{12} + R_3} \\
 u_{C,p} &= U_q \cdot \frac{R_3}{R_{12} + R_3} && C_1 \text{ corresponds to idle}
 \end{aligned}$$

4. superposition and **5. Determine constant(s)**:

$$\begin{aligned}
 u_C &= u_{C,h} + u_{C,p} \\
 u_C(t=0) &= K \cdot e^{\cancel{0}} + U_q \cdot \frac{R_3}{R_{12} + R_3} \stackrel{!}{=} U_q && \text{(Initial condition)} \\
 \Rightarrow K &= U_q \cdot \left(1 - \frac{R_3}{R_{12} + R_3} \right) \\
 &= U_q \cdot \frac{R_{12}}{R_{12} + R_3} \\
 u_C &= U_q \cdot \left(1 - \frac{R_3}{R_{12} + R_3} \right) \cdot e^{-\frac{t}{\tau}} + U_q \cdot \frac{R_3}{R_{12} + R_3} && \left| : R_3 \right. \\
 i_3 &= U_q \cdot \left(\frac{1}{R_3} - \frac{1}{R_{12} + R_3} \right) \cdot e^{-\frac{t}{\tau}} + U_q \cdot \frac{1}{R_{12} + R_3}
 \end{aligned}$$

Time constant τ and initial and final values of the current i_3 for sketch:

$$\begin{aligned}
 R_{12} &= R_1 || R_2 = \frac{1,2 \text{ k}\Omega \cdot 2 \text{ k}\Omega}{1,2 \text{ k}\Omega + 2 \text{ k}\Omega} = 750 \Omega \\
 \tau &= C_1 \cdot \frac{R_{12} \cdot R_3}{R_{12} + R_3} \\
 &= 1 \mu\text{F} \cdot \frac{750 \Omega \cdot 500 \Omega}{750 \Omega + 500 \Omega} = 0,3 \text{ ms} \\
 i_3(t=0_-) &= \frac{U_q}{R_2 + R_3} = \frac{250 \text{ V}}{2 \text{ k}\Omega + 500 \Omega} = 0,1 \text{ A} \\
 i_3(t=0_+) &= \frac{U_q}{R_3} = \frac{250 \text{ V}}{500 \Omega} = 0,5 \text{ A} \\
 i_3(t \rightarrow \infty) &= \frac{U_q}{R_{12} + R_3} = \frac{250 \text{ V}}{750 \Omega + 500 \Omega} = 0,2 \text{ A}
 \end{aligned}$$



B.6 Switching process for alternating voltage

a) Steady state ($t \rightarrow \infty$) with complex alternating current calculation:

$$\begin{aligned}
 \underline{I}_{L,e} &= \frac{\underline{U}_L}{\underline{Z}_L} = \frac{\underline{U}_1}{\underline{Z}_L} \cdot \frac{R_2 \parallel \underline{Z}_L}{R_1 + R_2 \parallel \underline{Z}_L} \quad \text{with } \underline{U}_1 = \hat{U}_1 \cdot e^{j0} = \hat{U}_1 \\
 &= \frac{\hat{U}_1}{\underline{Z}_L} \cdot \frac{\frac{R_2 \cdot \underline{Z}_L}{R_2 + \underline{Z}_L}}{R_2 + \frac{R_1 \cdot \underline{Z}_L}{R_2 + \underline{Z}_L}} \quad \left| \cdot \frac{R_2 + \underline{Z}_L}{R_2 + \underline{Z}_L} \right. \\
 &= \frac{\hat{U}_1}{\underline{Z}_L} \cdot \frac{R_2 \cdot \cancel{\underline{Z}_L}}{R_1 \cdot (R_2 + \underline{Z}_L) + R_2 \cdot \underline{Z}_L} \\
 &= \hat{U}_1 \cdot \frac{R_2}{R_1 \cdot R_2 + j\omega L \cdot (R_1 + R_2)} \\
 \varphi &= 0 + \cancel{\arctan\left(\frac{0}{R_2}\right)} - \arctan\left(\frac{\omega L \cdot (R_1 + R_2)}{R_1 \cdot R_2}\right) \\
 &= -\arctan\left(\frac{\omega L \cdot (R_1 + R_2)}{R_1 \cdot R_2}\right), \quad -\frac{\pi}{2} \leq \varphi \leq 0 \\
 \hat{I}_{L,e} &= \hat{U}_1 \cdot \frac{R_2}{\sqrt{(R_1 \cdot R_2)^2 + (\omega L \cdot (R_1 + R_2))^2}} \\
 i_{L,e}(t) &= \hat{I}_{L,e} \cdot \sin(\omega t + \varphi) \quad \text{current lags } u_1 \text{ with } \varphi < 0
 \end{aligned}$$

b) ODE for $i_L(t)$ ($t \geq t_0$):

$$\begin{aligned}
 u_1 &= u_{R_1} + u_L \\
 &= R_1 \cdot (i_2 + i_L) + L \cdot \frac{d}{dt} i_L \\
 &= R_1 \cdot \left(\frac{u_L}{R_2} + i_L \right) + L \cdot \frac{d}{dt} i_L \\
 &= \frac{R_1}{R_2} \cdot L \cdot \frac{d}{dt} i_L + R_1 \cdot i_L + L \cdot \frac{d}{dt} i_L \\
 &= L \cdot \left(\frac{R_1}{R_2} + 1 \right) \cdot \frac{d}{dt} i_L + R_1 \cdot i_L \quad \left| \cdot \frac{1}{R_1} \right. \\
 \frac{u_1}{R_1} &= L \cdot \underbrace{\left(\frac{1}{R_1} + \frac{1}{R_2} \right)}_{=\tau} \cdot \frac{d}{dt} i_L + i_L \\
 \tau &= L \cdot \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{L}{R_1 \parallel R_2}
 \end{aligned}$$

c) Time points for minimum and maximum volatile state (t_{min}, t_{max}).
 No settling if $i_{L,f} = 0 \Leftrightarrow i_L = i_{L,e} \Leftrightarrow i_L(t_0) = i_{L,e}(t_0)$ applies.

That is, for a vanishing transient state (no settling process), the following applies:

$$\begin{aligned}
 \text{IC: } i_L(t_0) &= i_{L,e}(t_0) = \hat{I}_{L,e} \cdot \sin(\omega t_0 + \varphi) \stackrel{!}{=} 0 \quad \Rightarrow i_L(t') = i_{L,e}(t') \Leftrightarrow i_{L,f}(t') = 0 \\
 &\Leftrightarrow \omega t_0 + \varphi \stackrel{!}{=} 0 + n \cdot \pi \quad \text{with } n \in \mathbb{N} \\
 t_{min} &= \frac{n \cdot \pi - \varphi}{\omega} \quad \text{corresponds to zero crossings of } i_{L,e}
 \end{aligned}$$

And for maximum transient state (switching point offset by 90°), the following applies:

$$t_{max} = \frac{n \cdot \pi - \varphi}{\omega} + \frac{T}{4} \quad \text{corresponds to extreme points of } i_{L,e}$$

- d) Maximum current during the transient process sought. Switching time t_0 at negative peak value of $i_{L,e}$. Let $t' = t - t_0$, i.e. $t' = 0$ at the switching time, then the following applies:

$$\begin{aligned} i_L(t') &= i_{L,h}(t') + i_{L,e}(t') \\ &= K \cdot e^{-\frac{t'}{\tau}} - \hat{I}_{L,e} \cdot \cos(\omega t') \\ \text{AB: } i_L(t' = 0) &= K - \hat{I}_{L,e} \stackrel{!}{=} 0 \quad \Rightarrow \quad K = \hat{I}_{L,e} \\ i_L(t') &= \hat{I}_{L,e} \cdot \left(e^{-\frac{t'}{\tau}} - \cos(\omega t') \right) \end{aligned}$$

Maximum at positive peak value (180° to t_0):

$$\begin{aligned} i_{L,max} &= i_L(t' = \frac{\pi}{\omega}) \\ &= \hat{I}_{L,e} \cdot \left(e^{-\frac{T/2}{T}} - \cos(\pi) \right) \\ &= \hat{I}_{L,e} \cdot \left(e^{-\frac{1}{2}} + 1 \right) = 1,606 \cdot \hat{I}_{L,e} \end{aligned}$$

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